

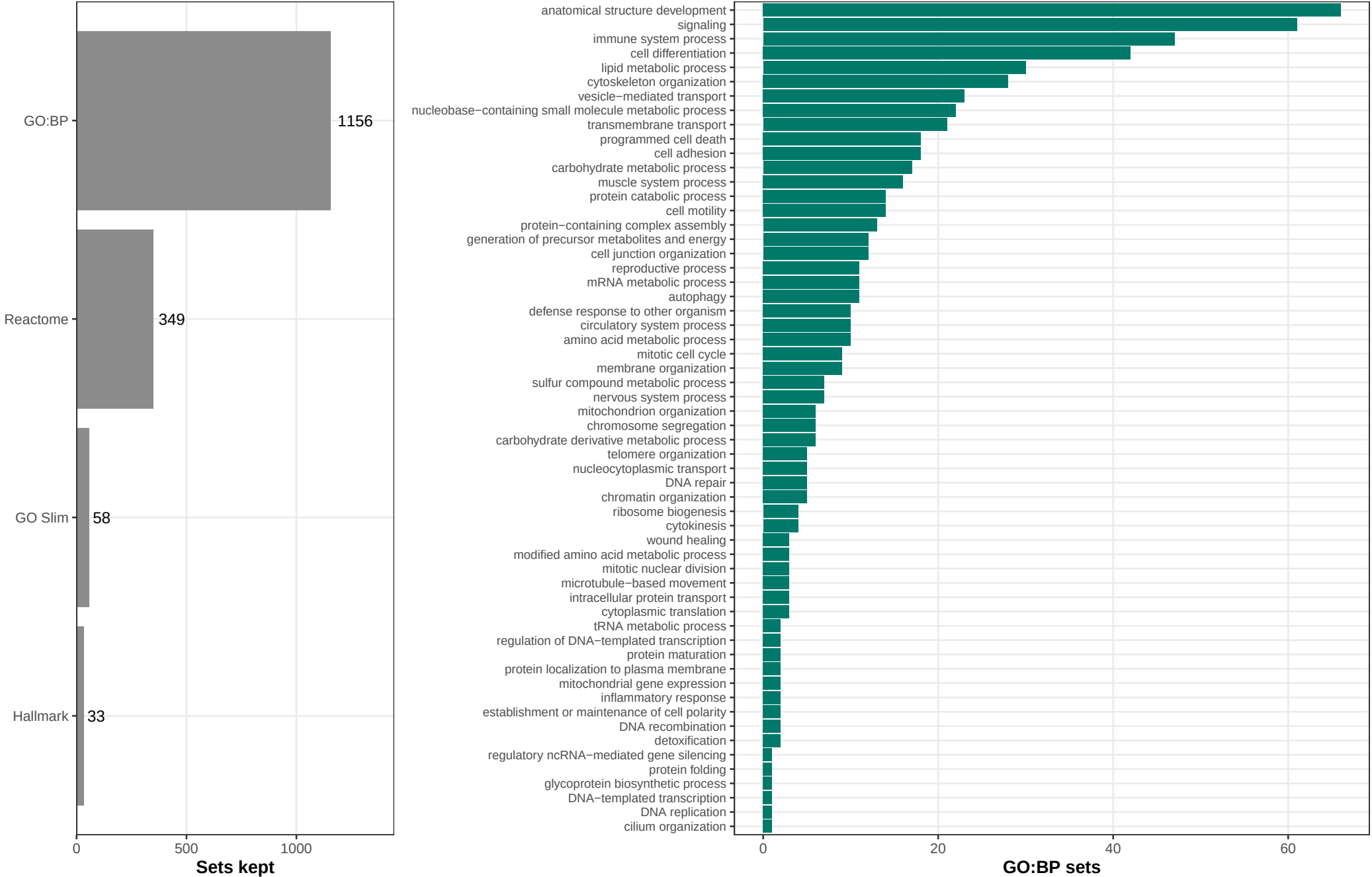
HRvLR 03 Pathways

Generated 2026-09-23

- 2 Gene-set universe and GO-Slim themes
- 3 fry set tests per contrast and collection
- 4 Pathway contrast hits (fry) (1/7)
- 5 Pathway contrast hits (fry) (2/7)
- 6 Pathway contrast hits (fry) (3/7)
- 7 Pathway contrast hits (fry) (4/7)
- 8 Pathway contrast hits (fry) (5/7)
- 9 Pathway contrast hits (fry) (6/7)
- 10 Pathway contrast hits (fry) (7/7)
- 11 Leading non-redundant sets by fgsea
- 12 Theme-level direction per contrast
- 13 Samples in pathway-score space
- 14 Pathway AUC per classification task
- 15 Pathway association with phenotype
- 16 Pathway classification hits (1/5)
- 17 Pathway classification hits (2/5)
- 18 Pathway classification hits (3/5)
- 19 Pathway classification hits (4/5)
- 20 Pathway classification hits (5/5)
- 21 Pathway association hits, level at T1 (1/7)
- 22 Pathway association hits, level at T1 (2/7)
- 23 Pathway association hits, level at T1 (3/7)
- 24 Pathway association hits, level at T1 (4/7)
- 25 Pathway association hits, level at T1 (5/7)
- 26 Pathway association hits, level at T1 (6/7)
- 27 Pathway association hits, level at T1 (7/7)
- 28 Pathway association hits, training change (T2 – T1) (1/6)
- 29 Pathway association hits, training change (T2 – T1) (2/6)
- 30 Pathway association hits, training change (T2 – T1) (3/6)
- 31 Pathway association hits, training change (T2 – T1) (4/6)
- 32 Pathway association hits, training change (T2 – T1) (5/6)
- 33 Pathway association hits, training change (T2 – T1) (6/6)
- 34 Pathway association hits, acute change (T3 – T2) (1/5)
- 35 Pathway association hits, acute change (T3 – T2) (2/5)
- 36 Pathway association hits, acute change (T3 – T2) (3/5)
- 37 Pathway association hits, acute change (T3 – T2) (4/5)
- 38 Pathway association hits, acute change (T3 – T2) (5/5)

Gene-set universe and GO-Slim themes

MSigDB 2026.1.Hs and GO-Slim generic (2026-07-26); 15 to 500 detected members per set



Left: sets kept per collection after the detected-member floor. Right: GO:BP sets grouped by their most specific GO-Slim ancestor (657 of 1156 GO:BP sets fall under a slim term; the rest carry no theme). Data: 03_Pathways/01_Gene_Sets/c_data/gene_sets.xlsx, sheets catalog and themes.

fry set tests per contrast and collection

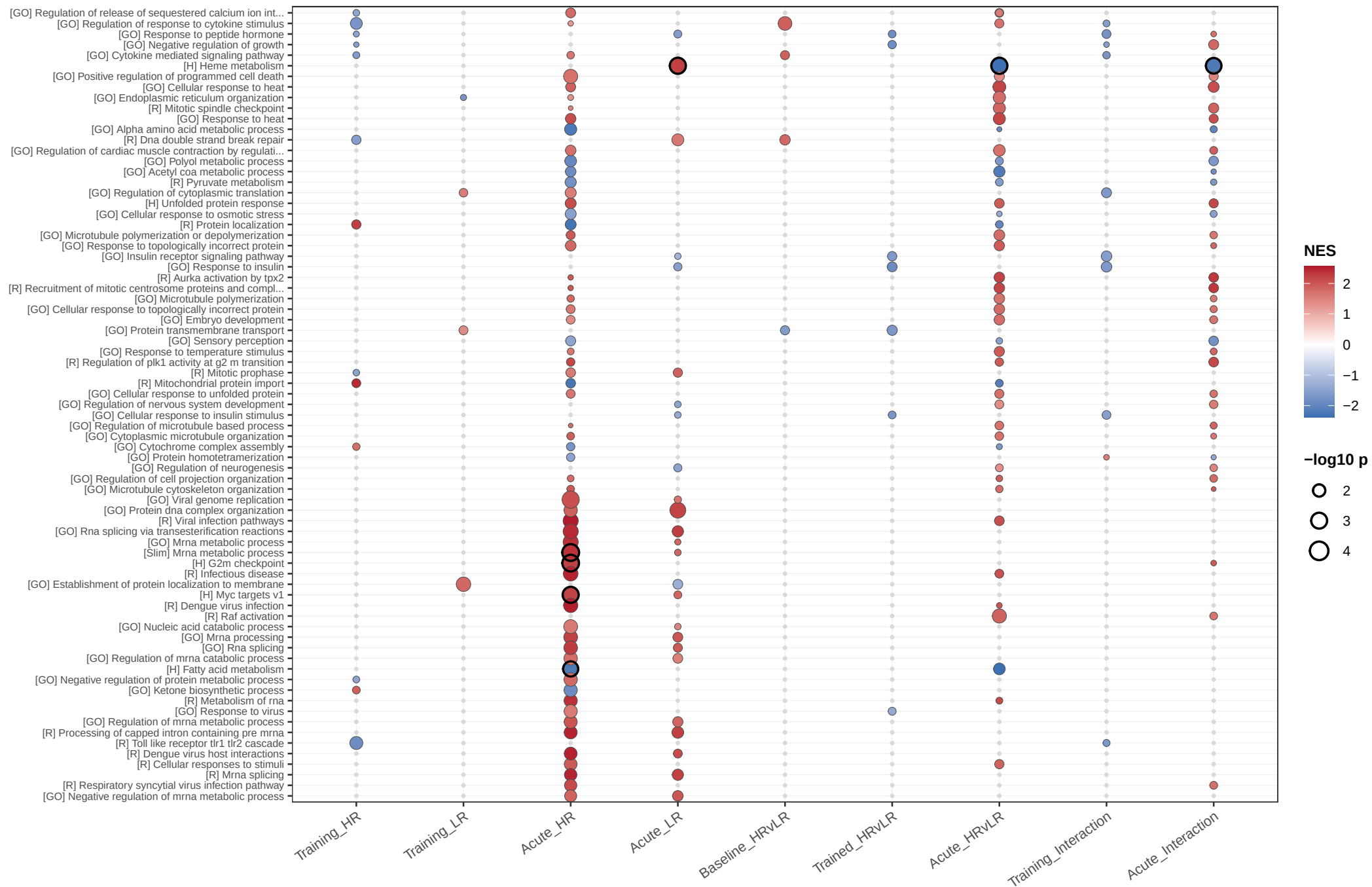
limma::fry, protein design, subject block, cor 0.176; BH per contrast x collection



Label: sets at nominal fry $p < 0.05$ / sets at BH < 0.05 . Fill: nominal sets over the count expected by chance ($0.05 \times$ sets in the collection); white is what no signal looks like. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx, sheet summary.

Pathway contrast hits (fry) (1/7)

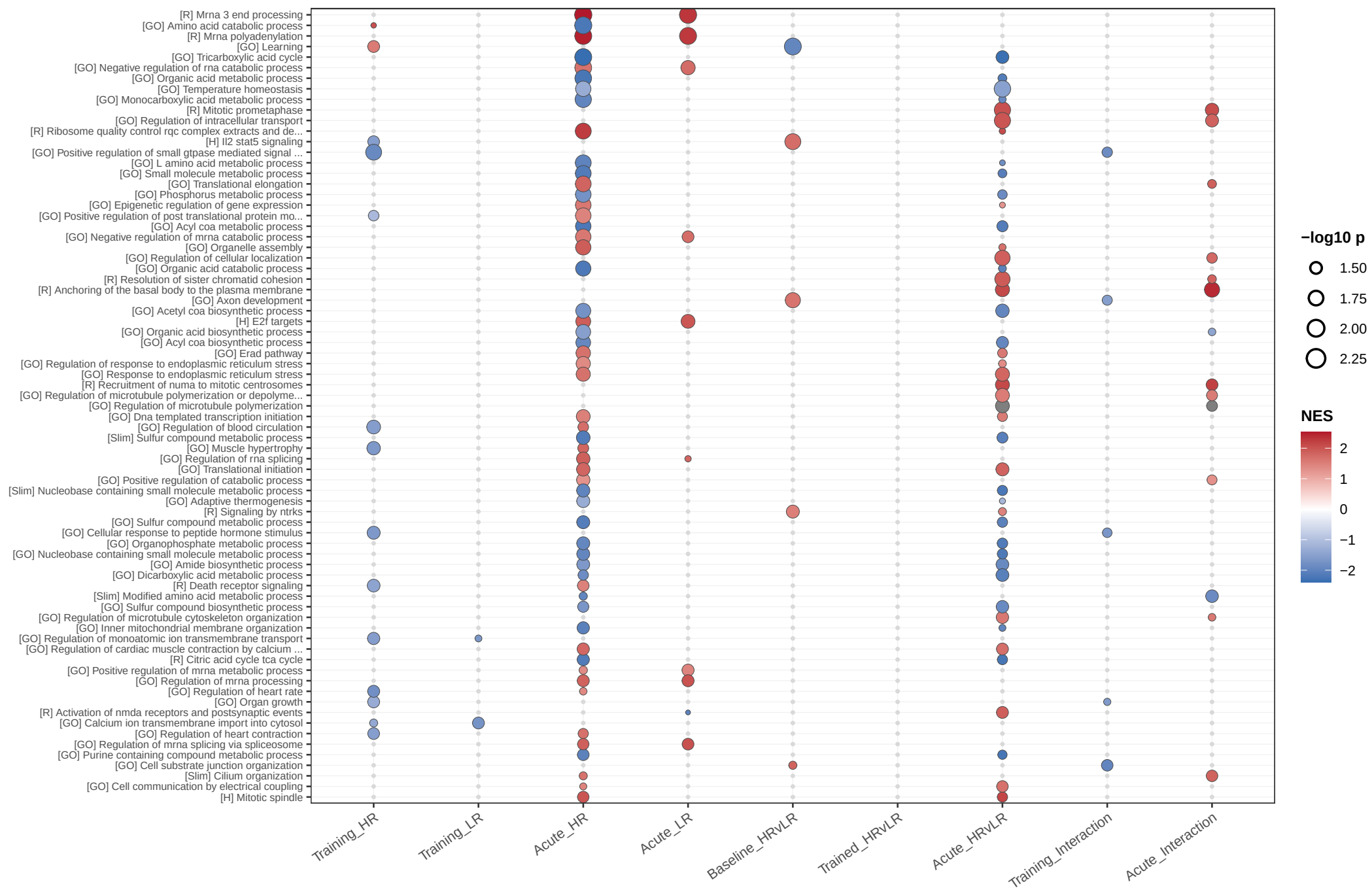
fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = NES, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

Pathway contrast hits (fry) (2/7)

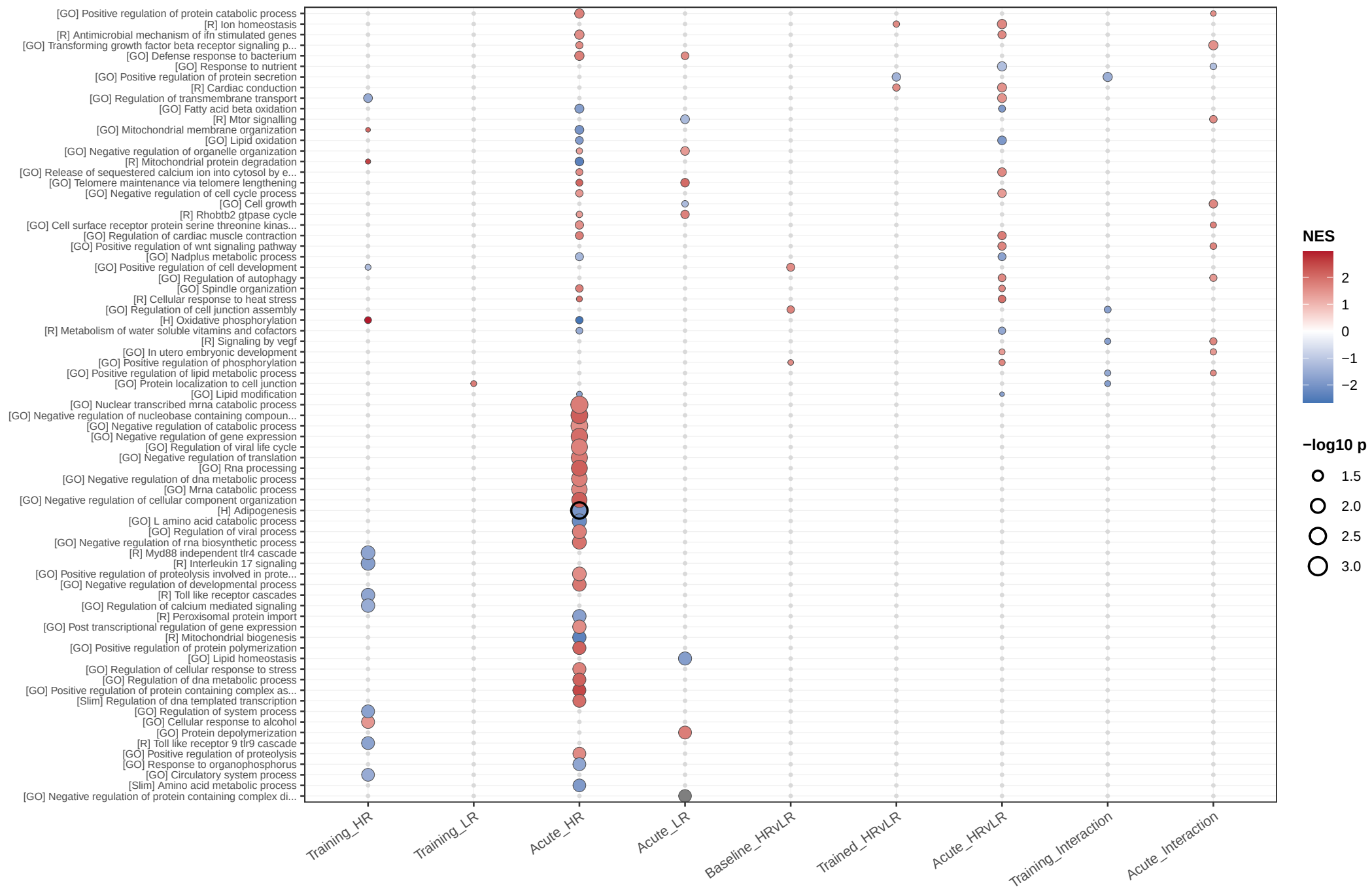
fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 76–150



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = NES, size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

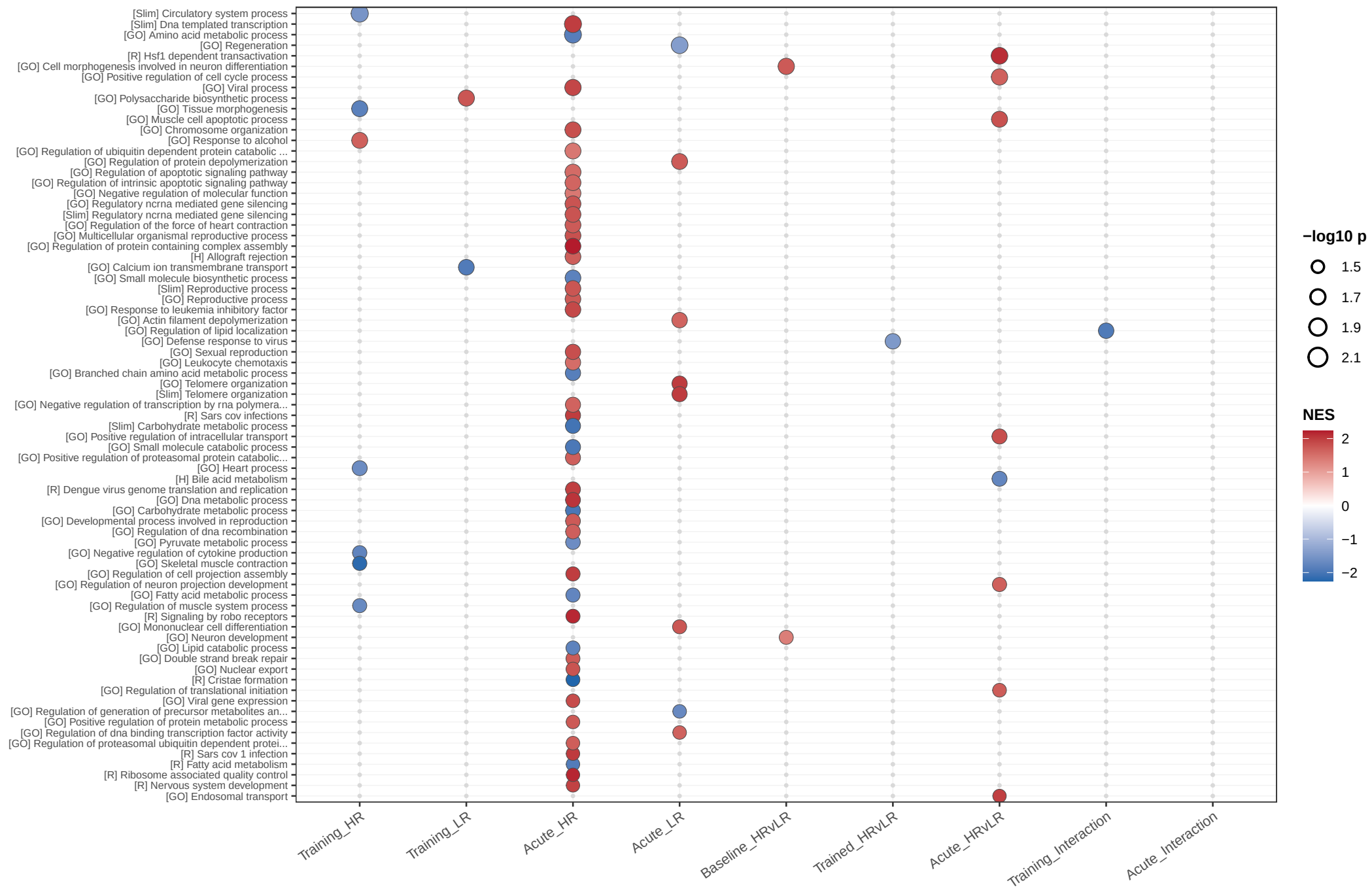
Pathway contrast hits (fry) (3/7)

fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 151–225



Pathway contrast hits (fry) (4/7)

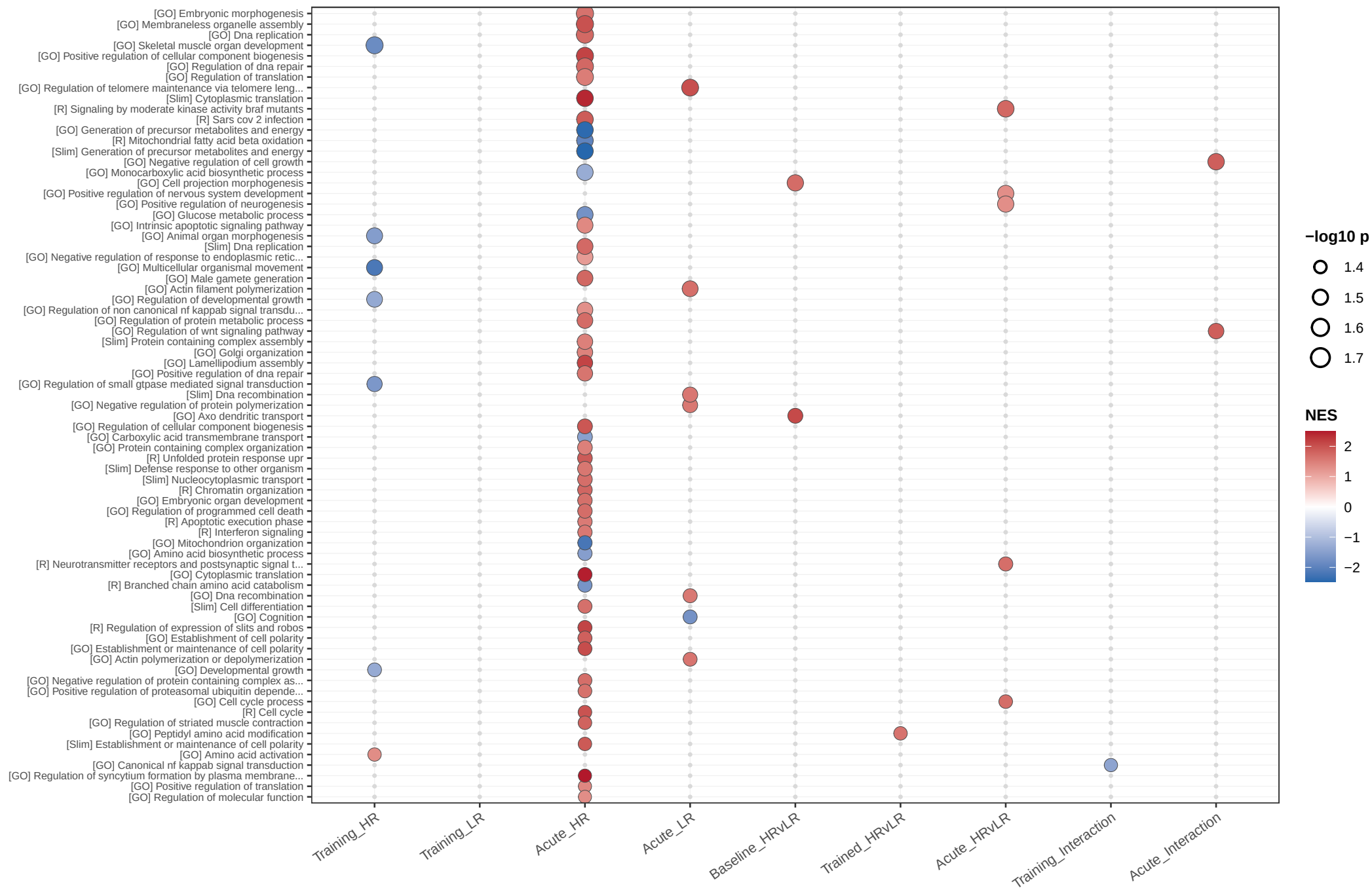
fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 226–300



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = NES, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

Pathway contrast hits (fry) (5/7)

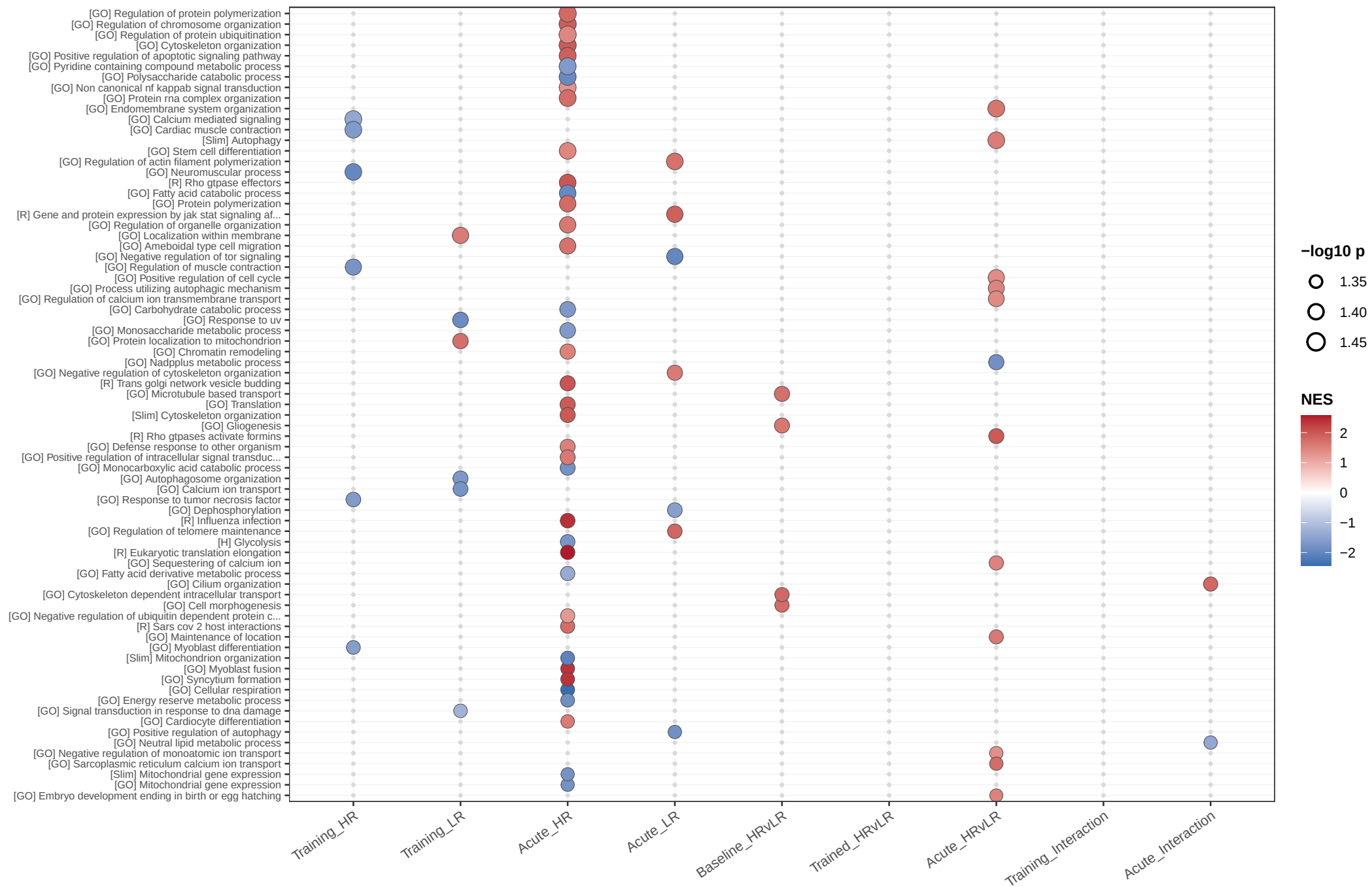
fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 301–375



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = NES, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

Pathway contrast hits (fry) (6/7)

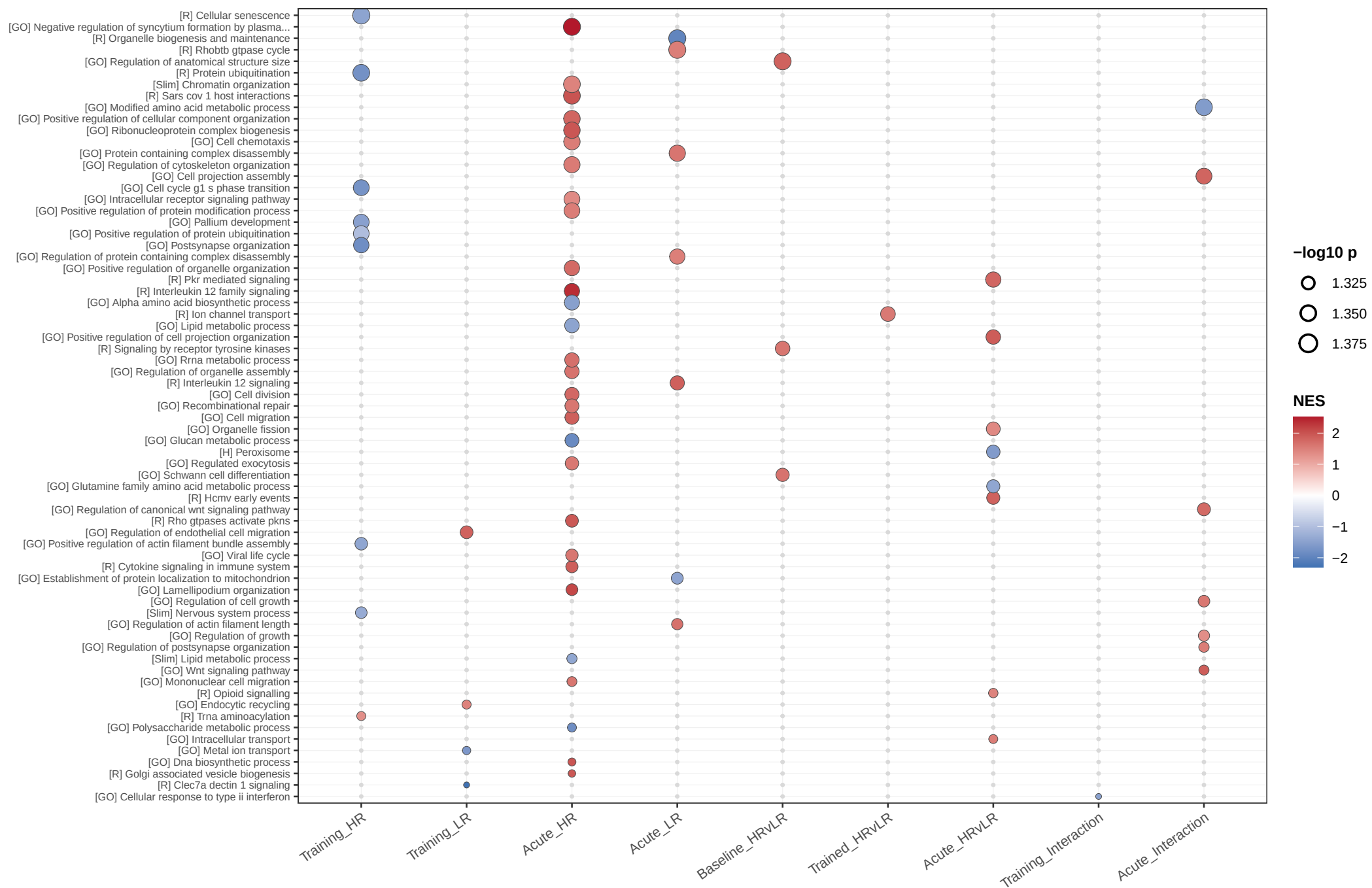
fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 376–450



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = NES, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

Pathway contrast hits (fry) (7/7)

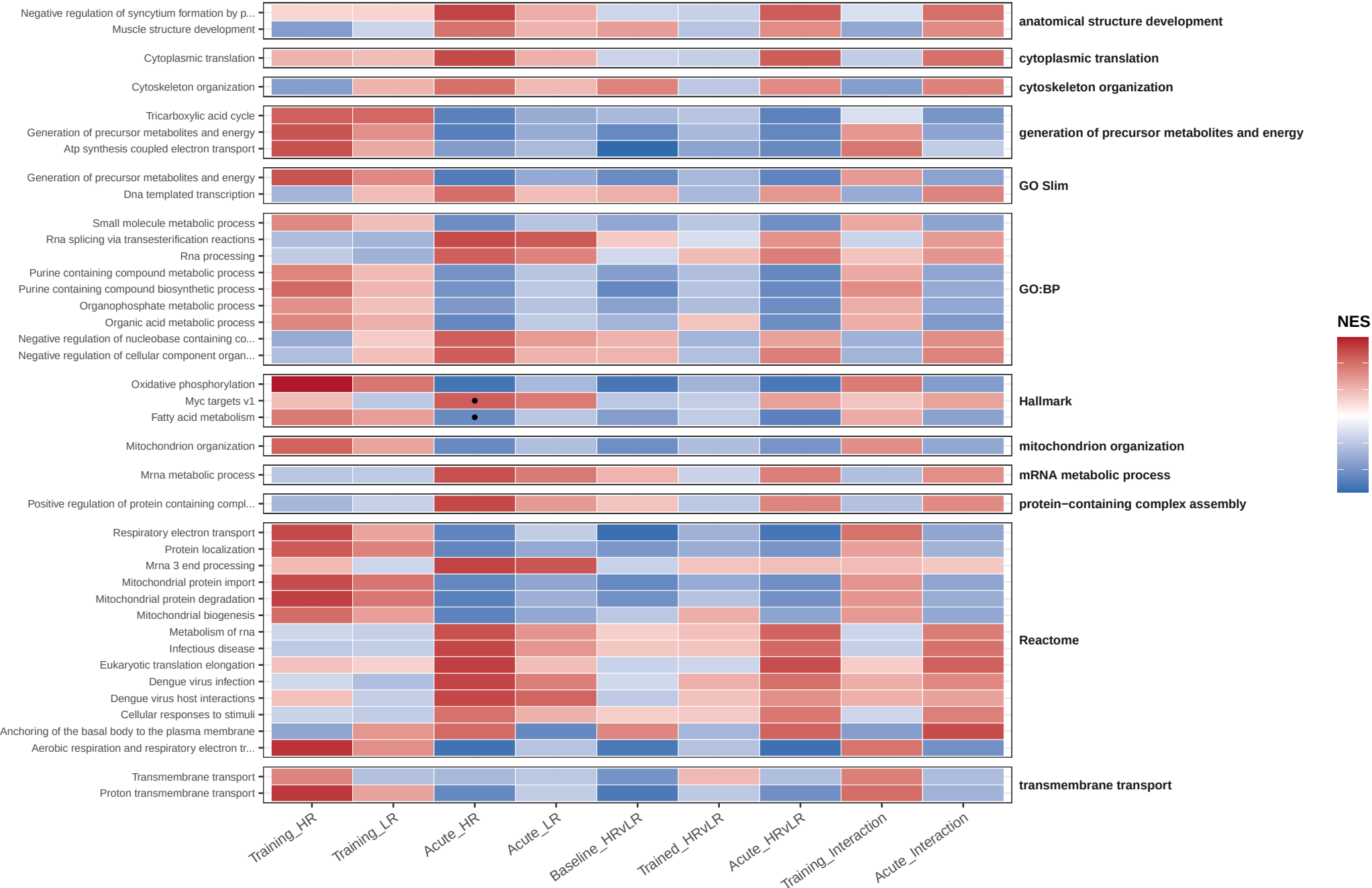
fry p per contrast, filled by fgsea NES for direction; 519 features nominal in at least one column; rows 451–519



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = NES, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

Leading non-redundant sets by fgsea

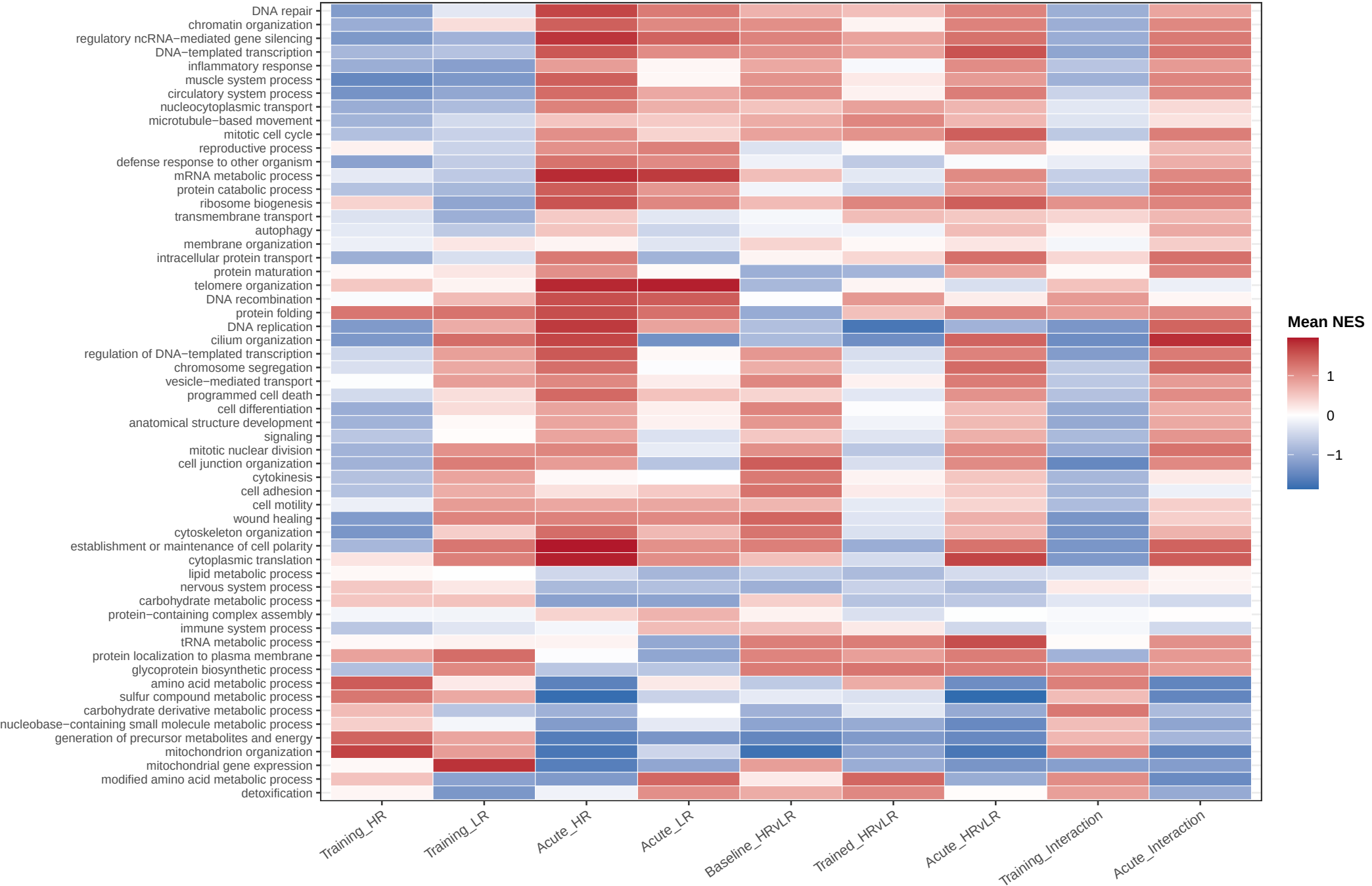
fgsea on moderated t; collapsePathways main sets, padj < 0.05; top 40



Rows: sets fgsea calls at padj < 0.05 in at least one contrast and that collapsePathways keeps as non-redundant, grouped by GO-Slim theme (or collection where a set has none). Fill: NES in every contrast. Dot: fry BH < 0.05 in that contrast. fgsea's gene-permutation p assumes exchangeable proteins, so fry is the test to quote. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

Theme-level direction per contrast

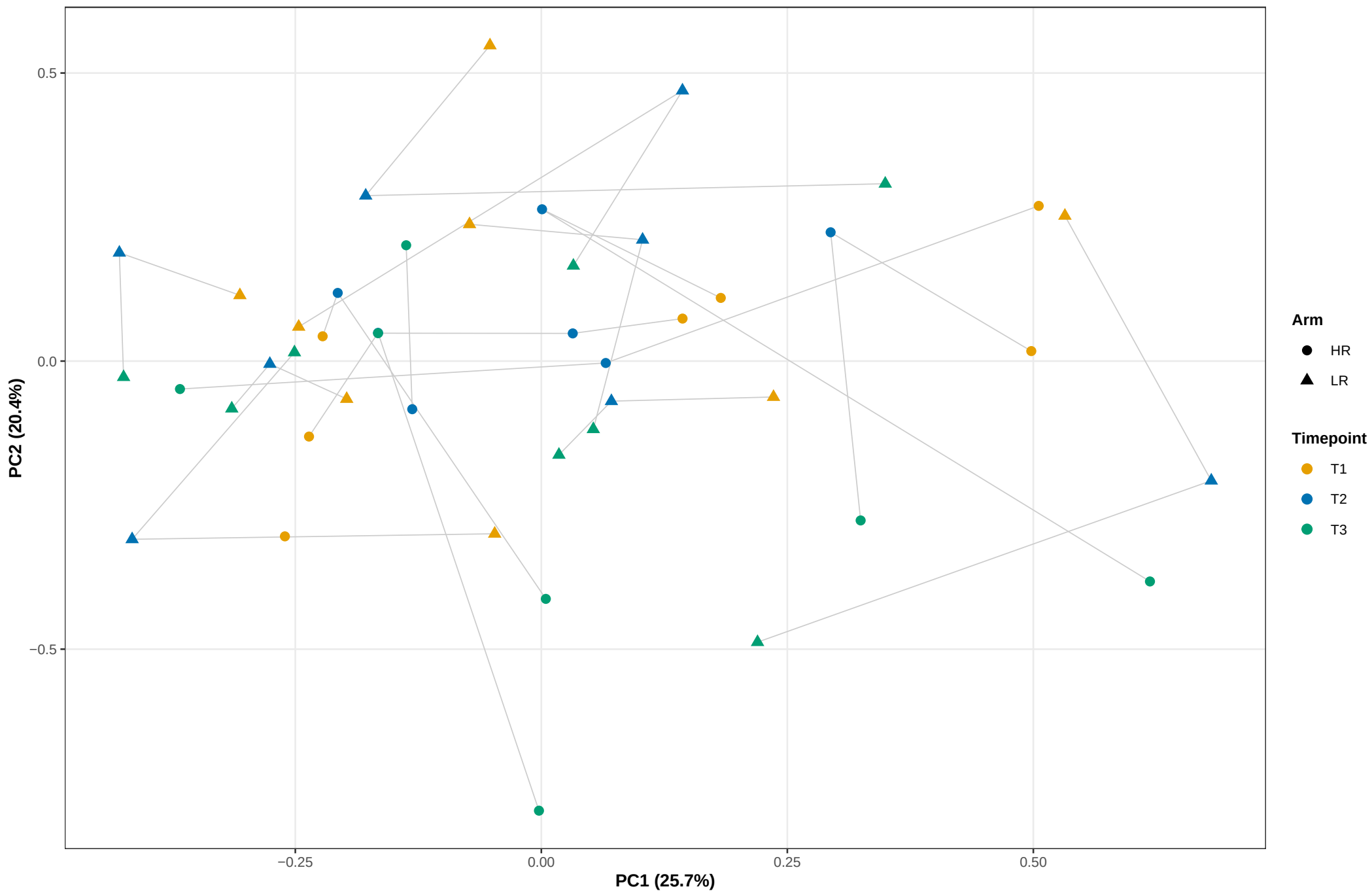
Mean fgsea NES of the GO:BP sets in each theme; rows clustered



Each tile averages the NES of every GO:BP set sharing a GO-Slim theme; it summarises direction, not significance, and a theme of few sets moves with any one of them. Sets per theme: page 2. Data: 03_Pathways/02_Set_Tests/c_data/set_tests.xlsx.

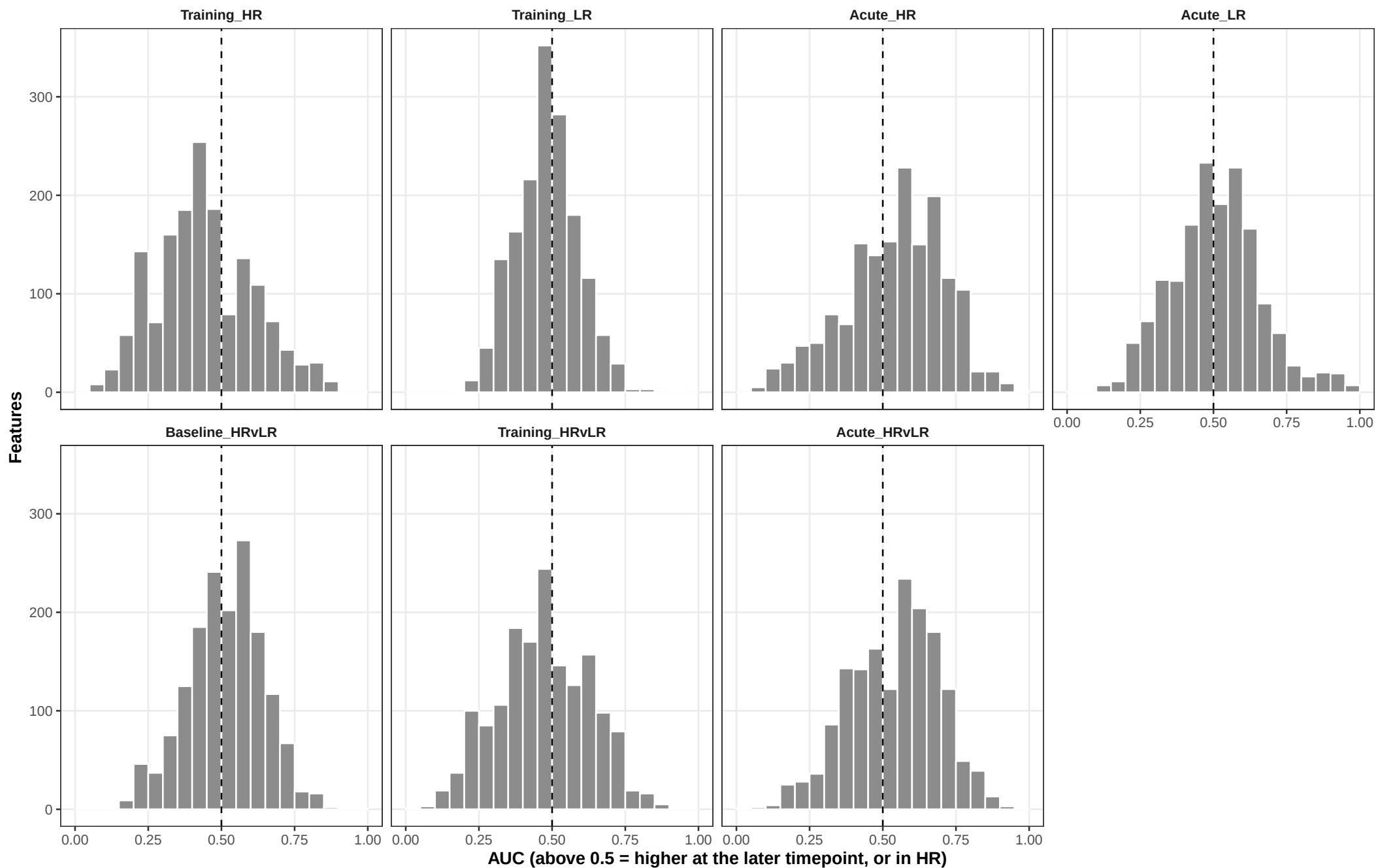
Samples in pathway-score space

PCA of 1596 singscore set scores x 45 samples



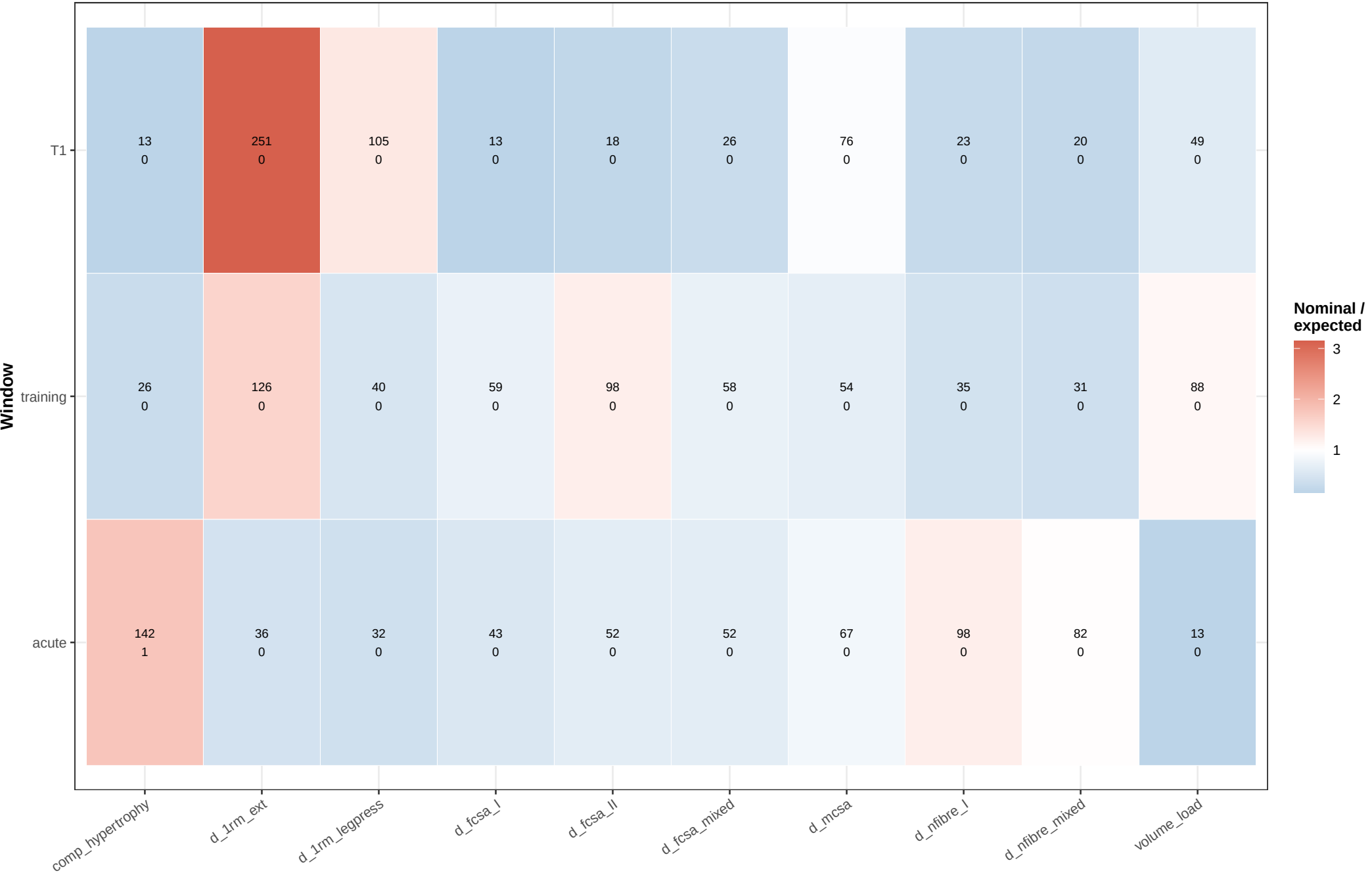
Pathway AUC per classification task

Mann-Whitney AUC, Wilcoxon p (signed-rank within arm); 1596 features



Each feature contributes one AUC per task. Within-arm tasks compare a subject's later timepoint with its earlier one; HRvLR tasks compare subjects on the baseline level or on their own change. Dashed line: chance. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheets classify and chance_classify.

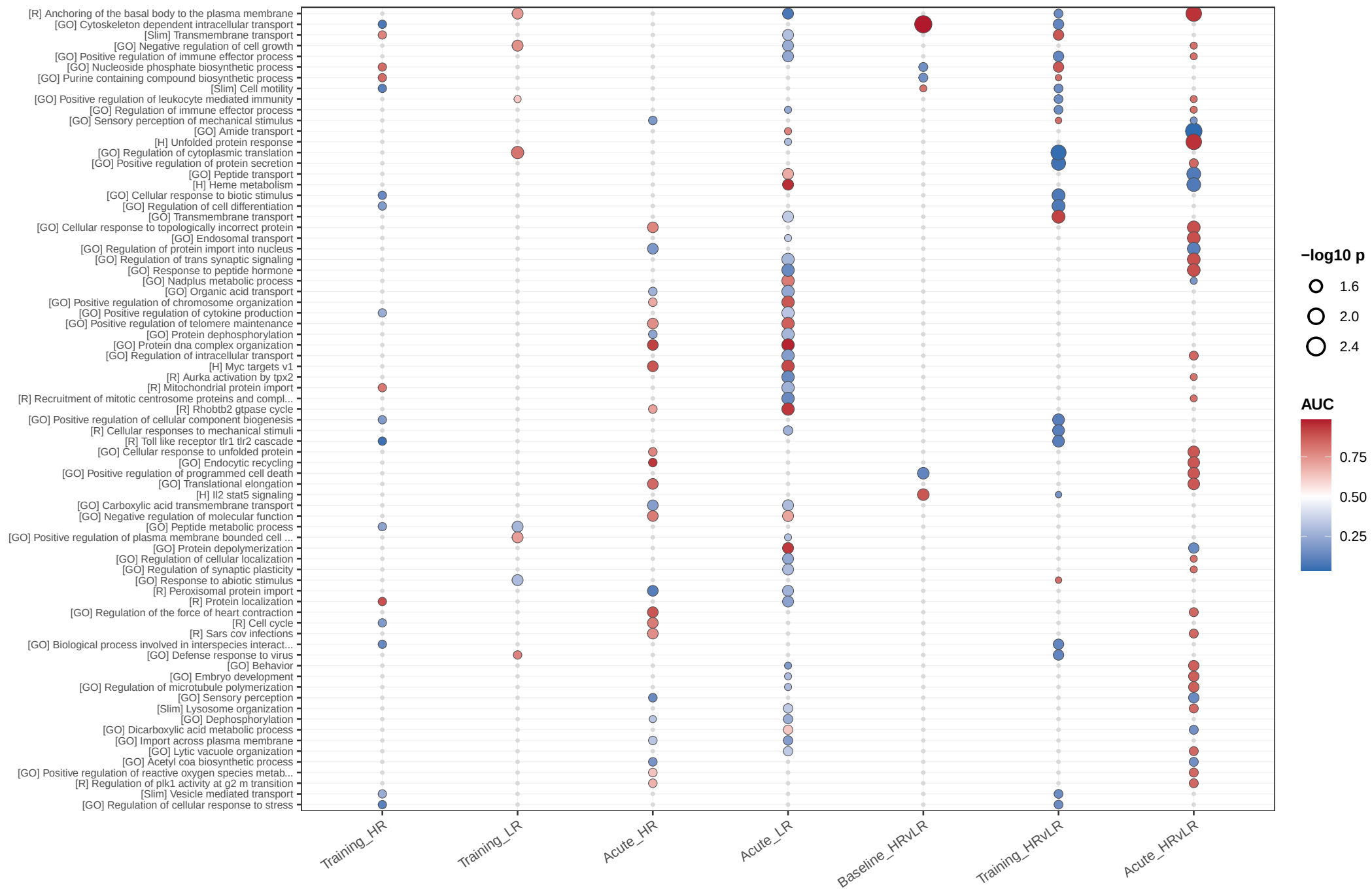
Pathway association with phenotype
limma on a continuous predictor; BH per cell; 1596 features



Rows: the feature at T1, or its change over training (T2 – T1) or the acute bout (T3 – T2). Columns: the ten phenotypes. Fill: nominal hits at $p < 0.05$ over the count expected by chance ($0.05 \times$ tested); white is what an empty screen looks like. Top number: nominal hits; bottom: BH < 0.05 . Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheets associate and chance_associate.

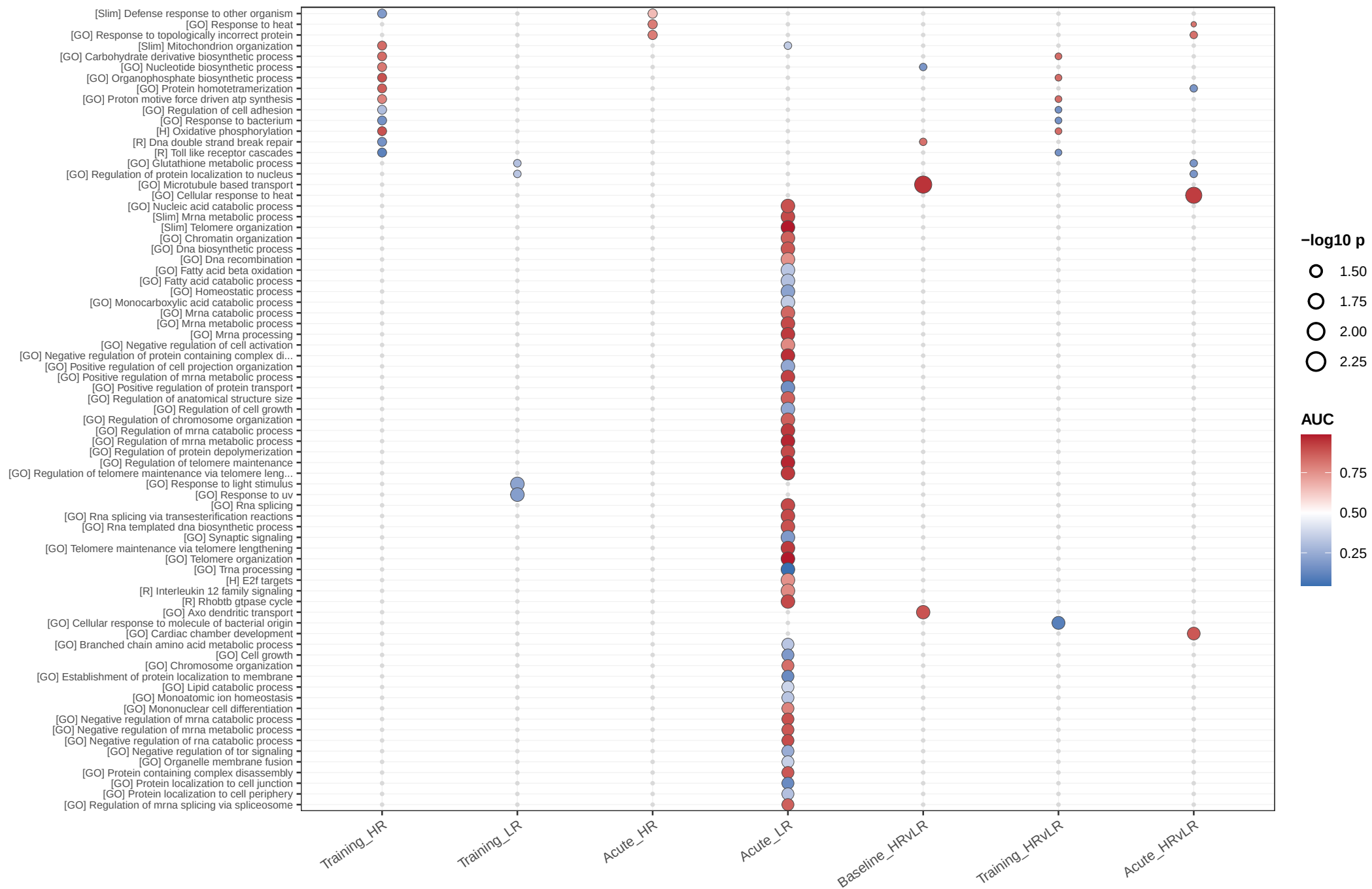
Pathway classification task hits (1/5)

AUC and Wilcoxon p per task, paired within arm; 374 features nominal in at least one column; rows 1–75



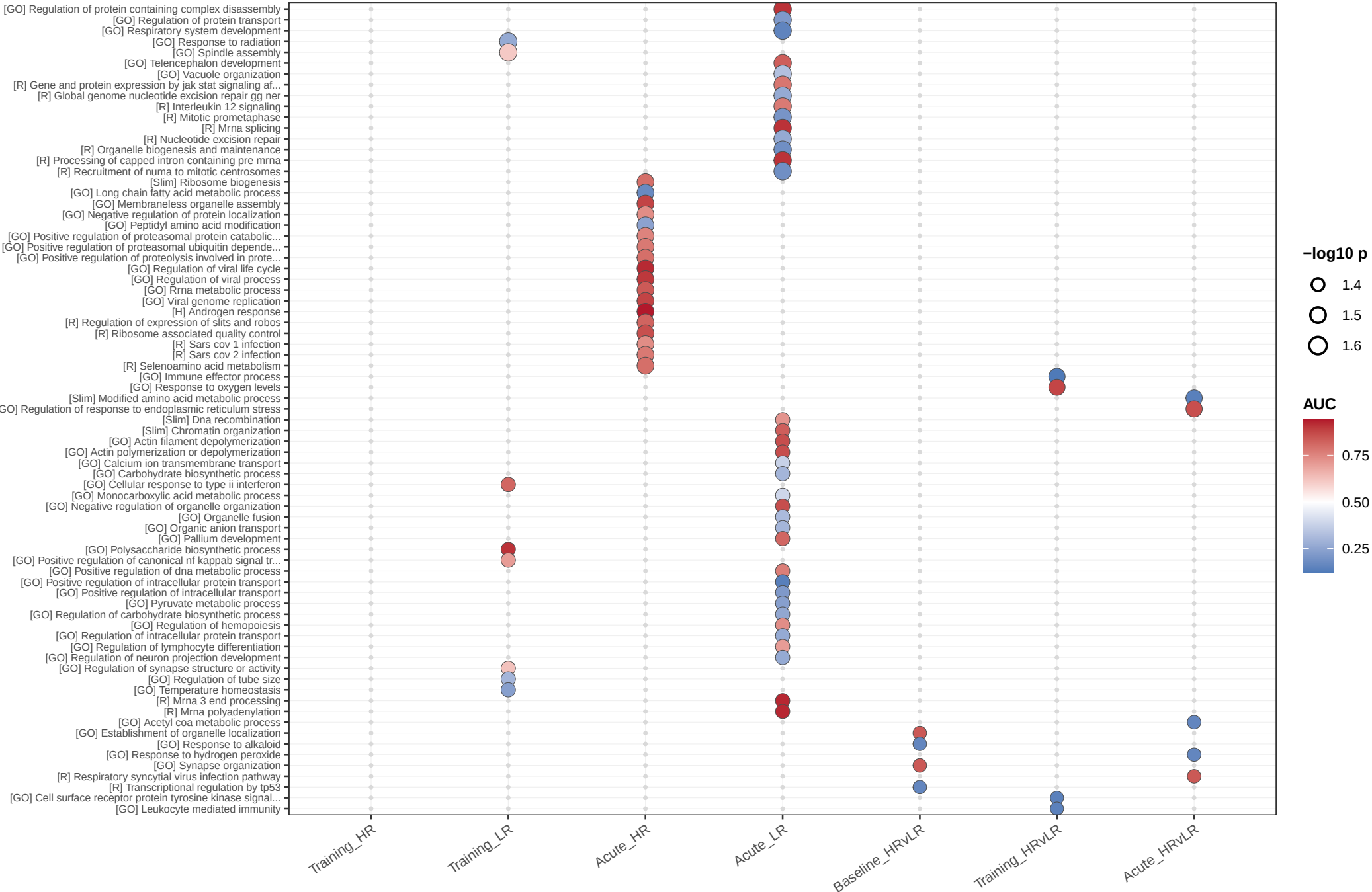
Pathway classification hits (2/5)

AUC and Wilcoxon p per task, paired within arm; 374 features nominal in at least one column; rows 76–150



Pathway classification hits (3/5)

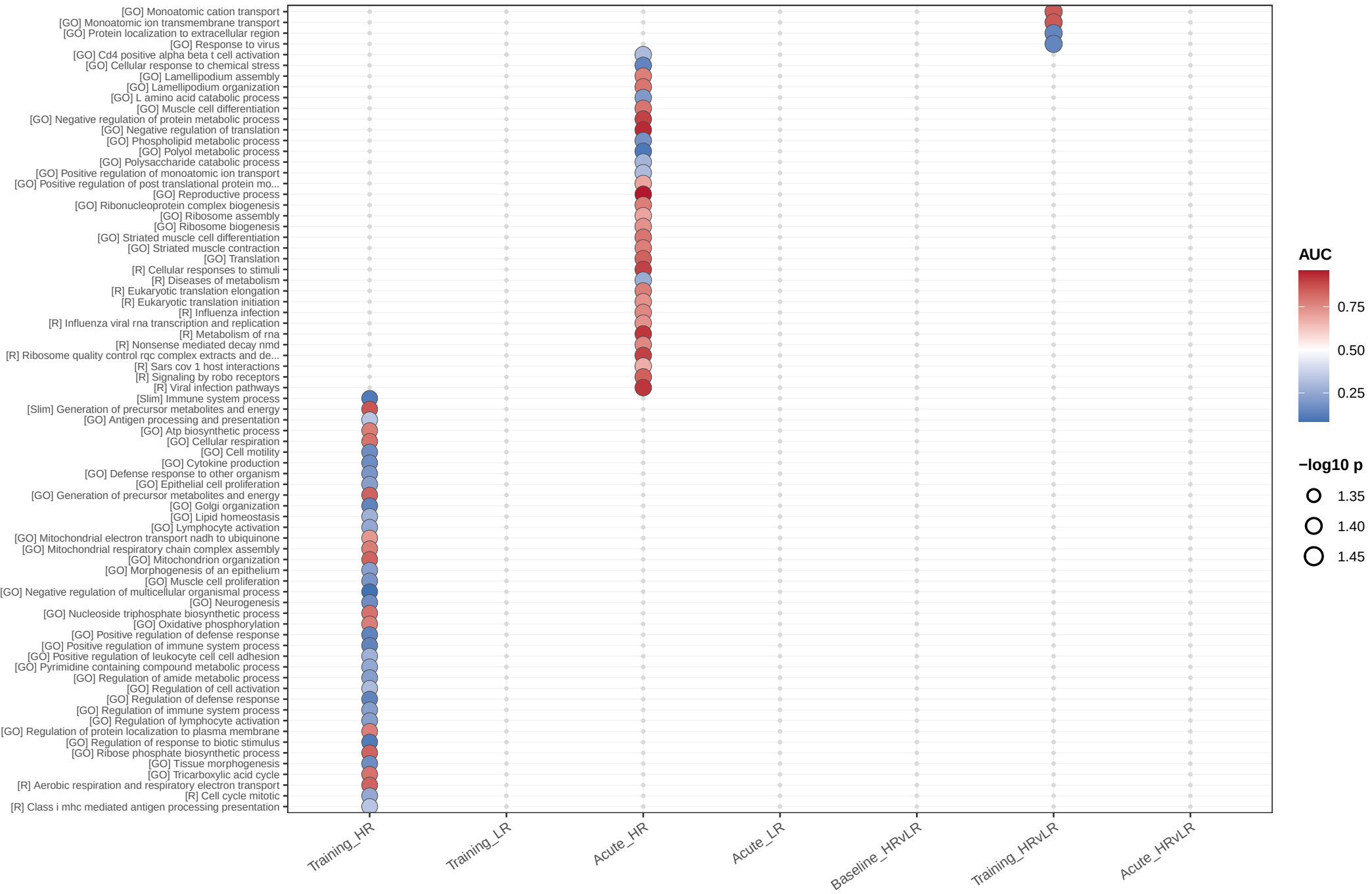
AUC and Wilcoxon p per task, paired within arm; 374 features nominal in at least one column; rows 151–225



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = AUC, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet classify.

Pathway classification hits (4/5)

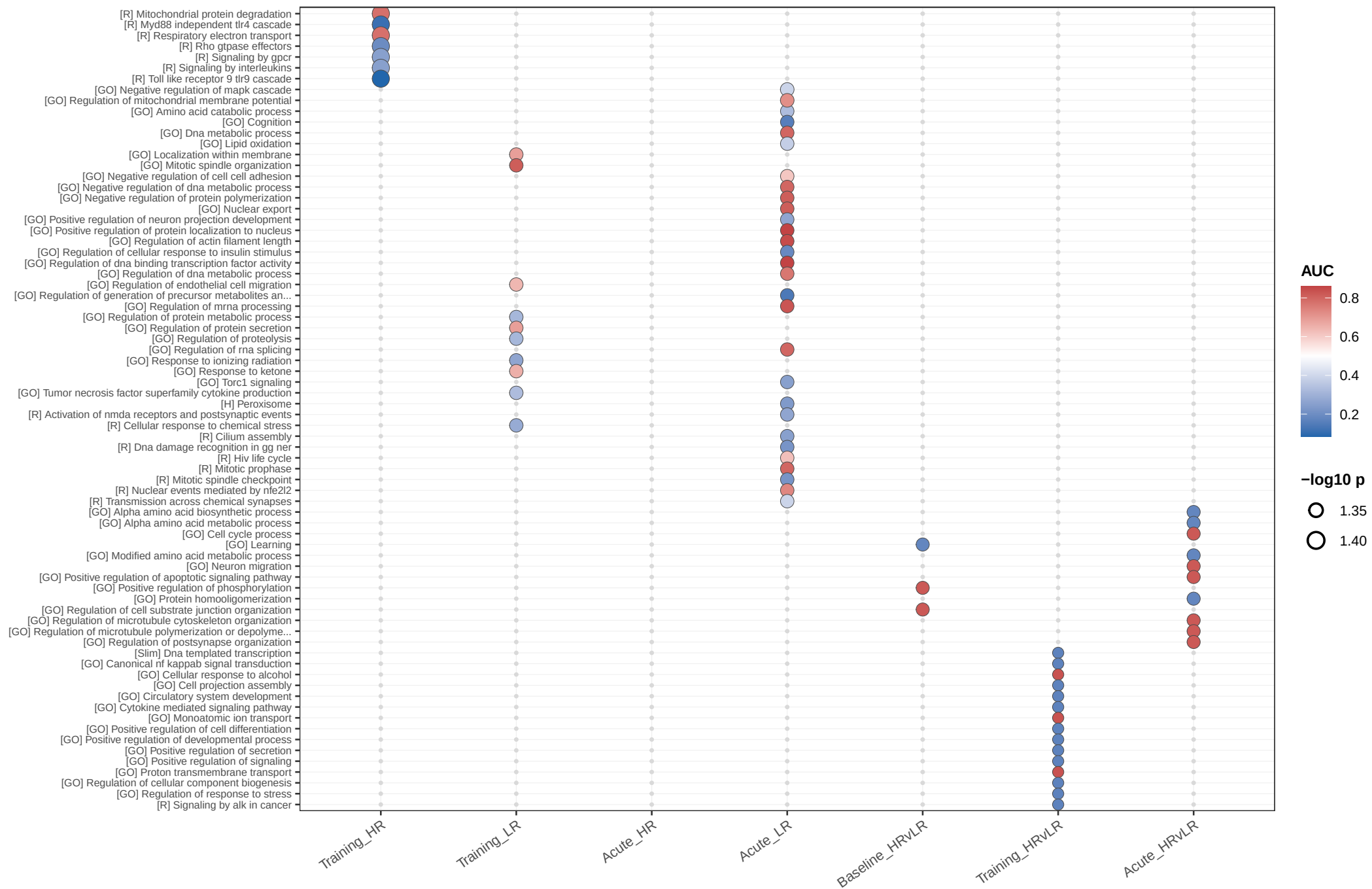
AUC and Wilcoxon p per task, paired within arm; 374 features nominal in at least one column; rows 226–300



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = AUC, size = –log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet classify.

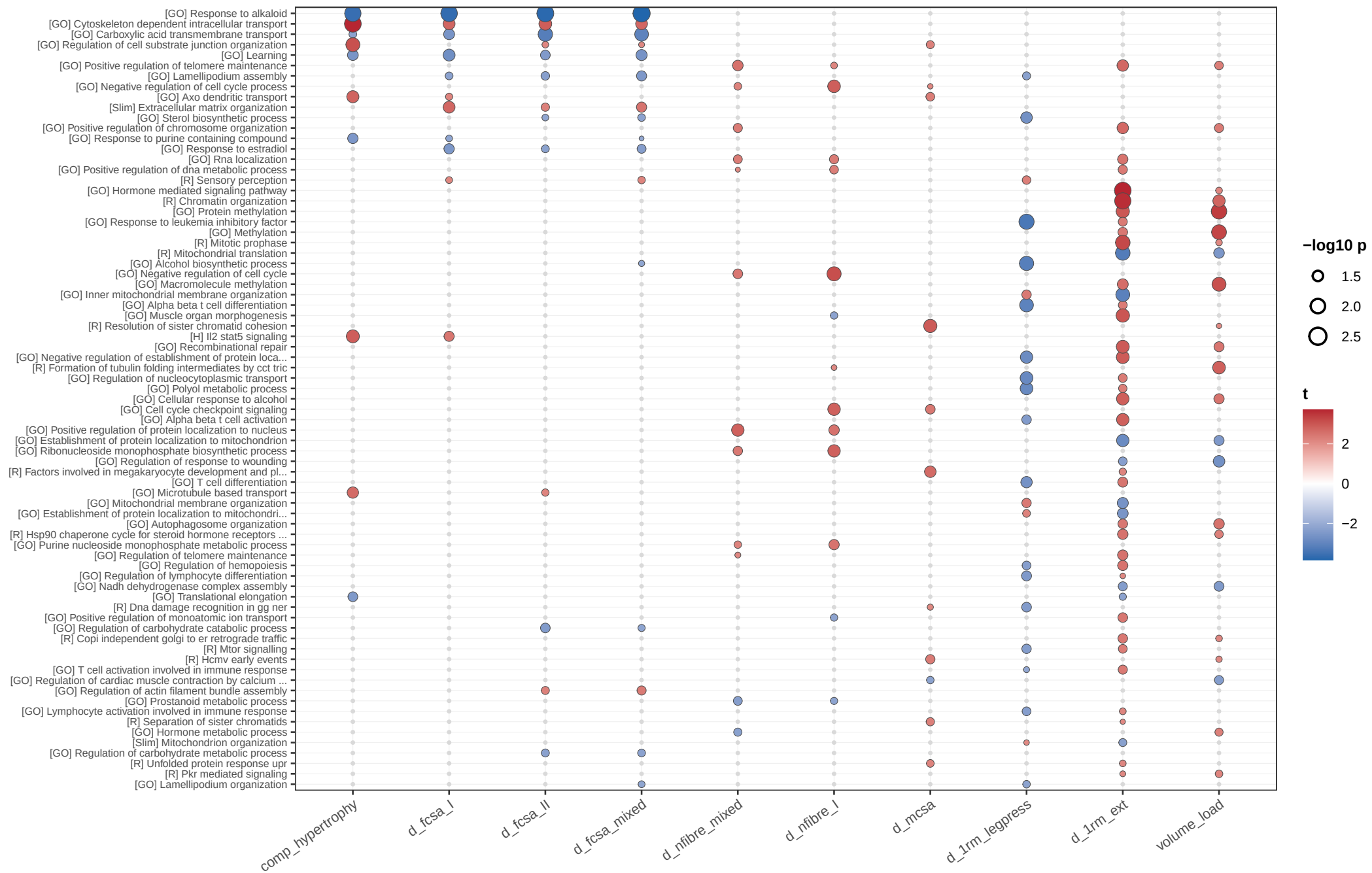
Pathway classification hits (5/5)

AUC and Wilcoxon p per task, paired within arm; 374 features nominal in at least one column; rows 301–374



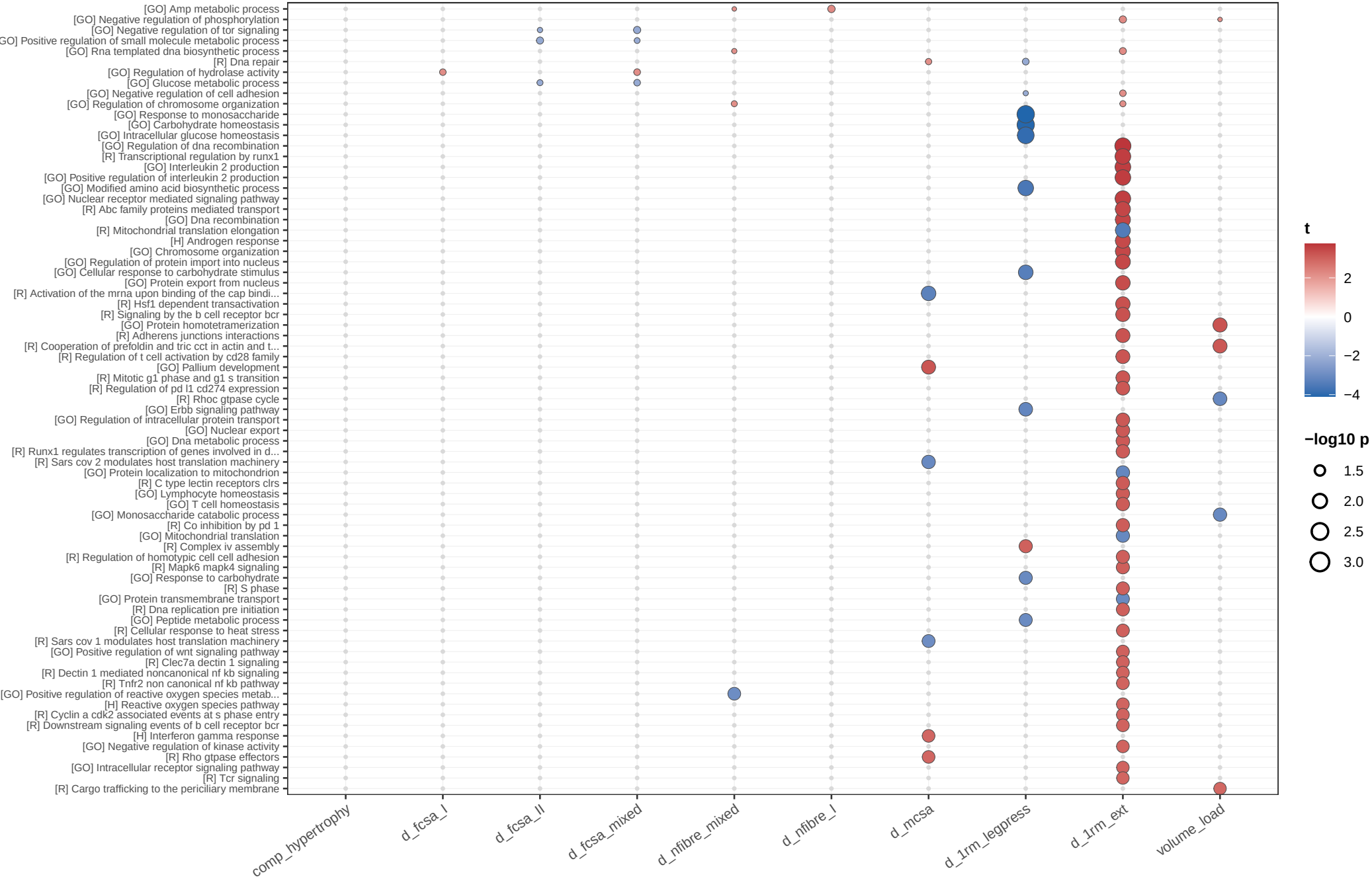
Pathway association hits, level at T1 (1/7)

limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: $BH < 0.05$. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

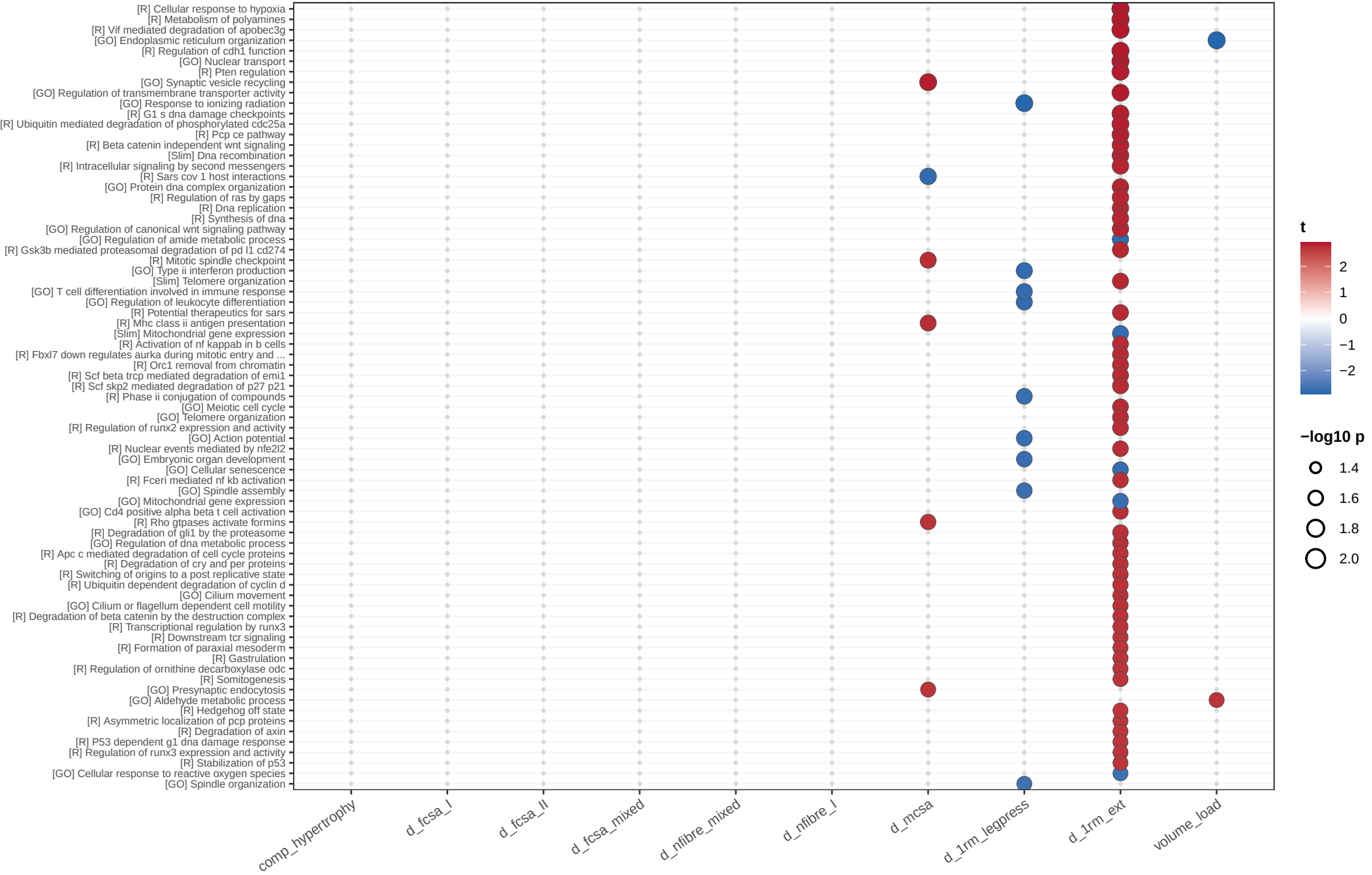
Pathway association hits, level at T1 (2/7)
 limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 76–150



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, level at T1 (3/7)

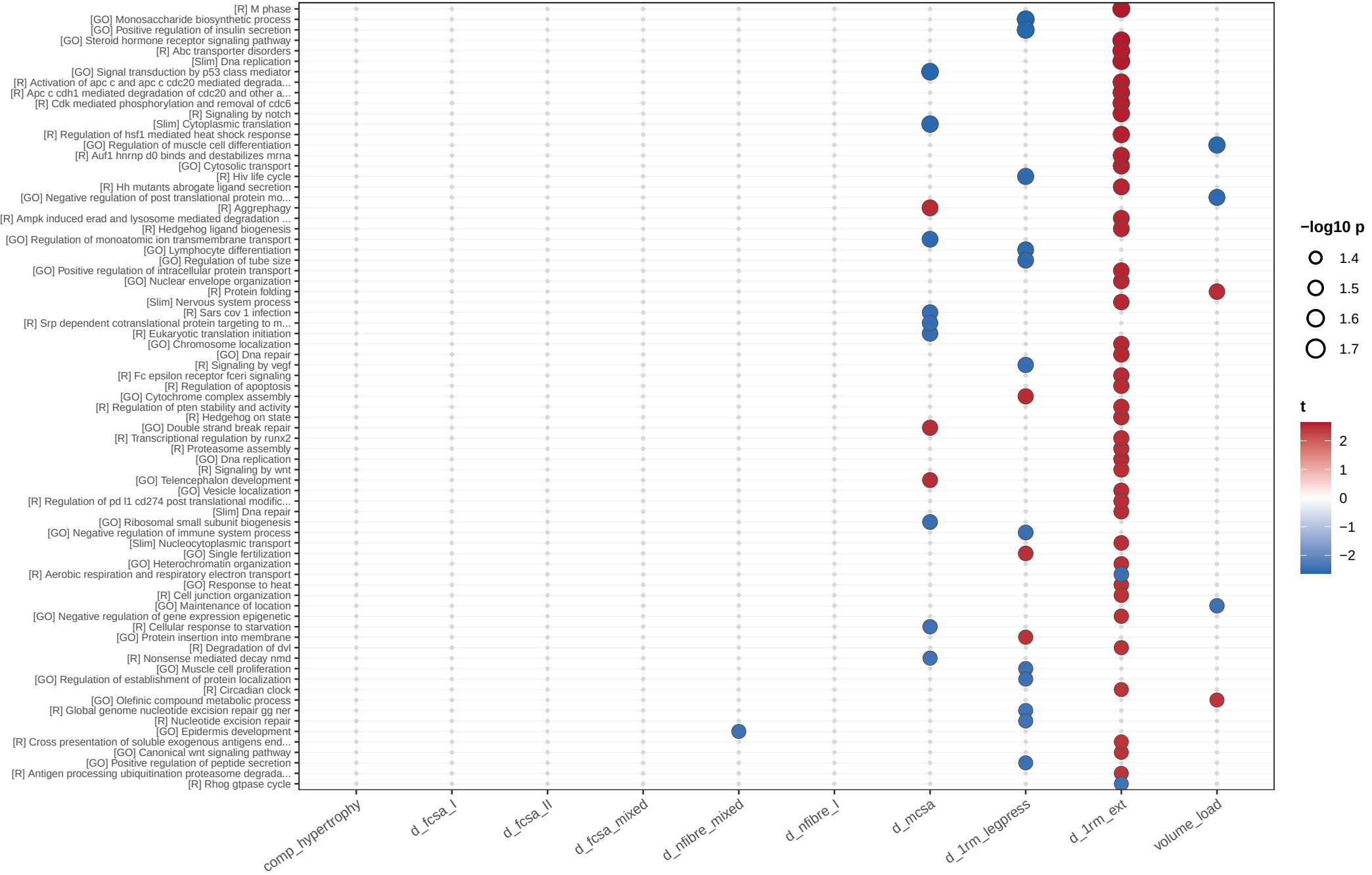
limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 151–225



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

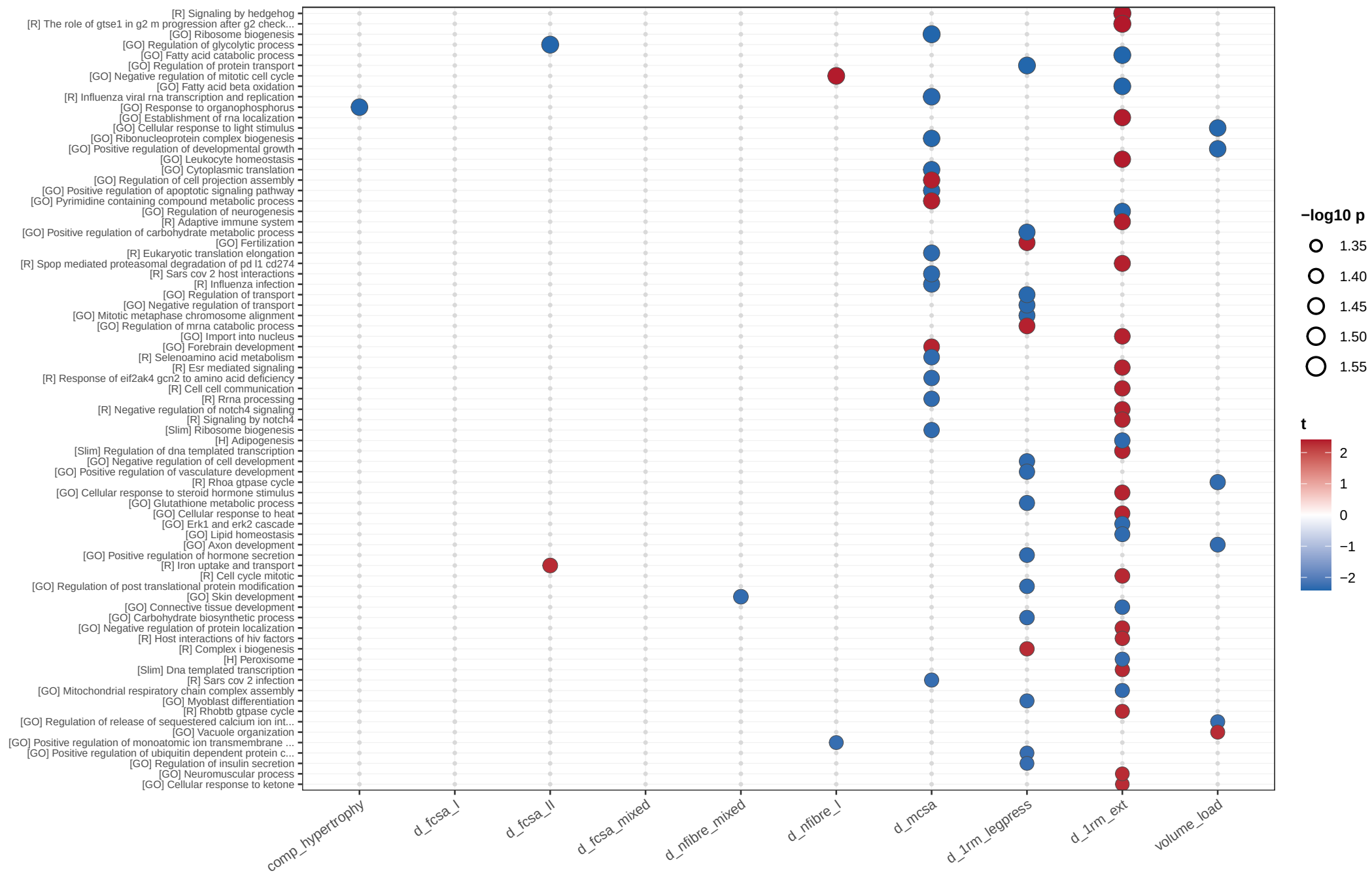
Pathway association hits, level at T1 (4/7)

limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 226–300



Pathway association hits, level at T1 (5/7)

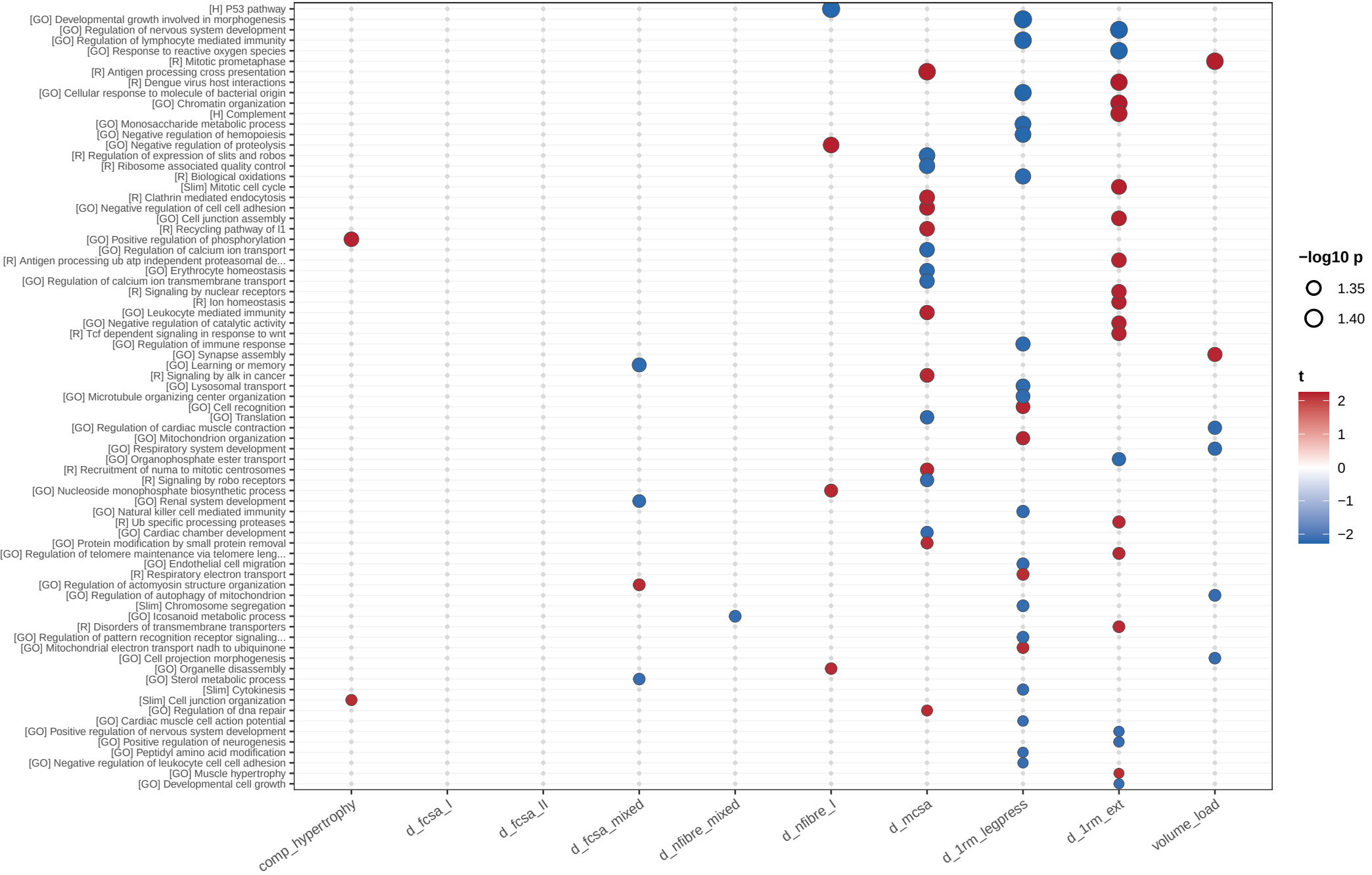
limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 301–375



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, level at T1 (6/7)

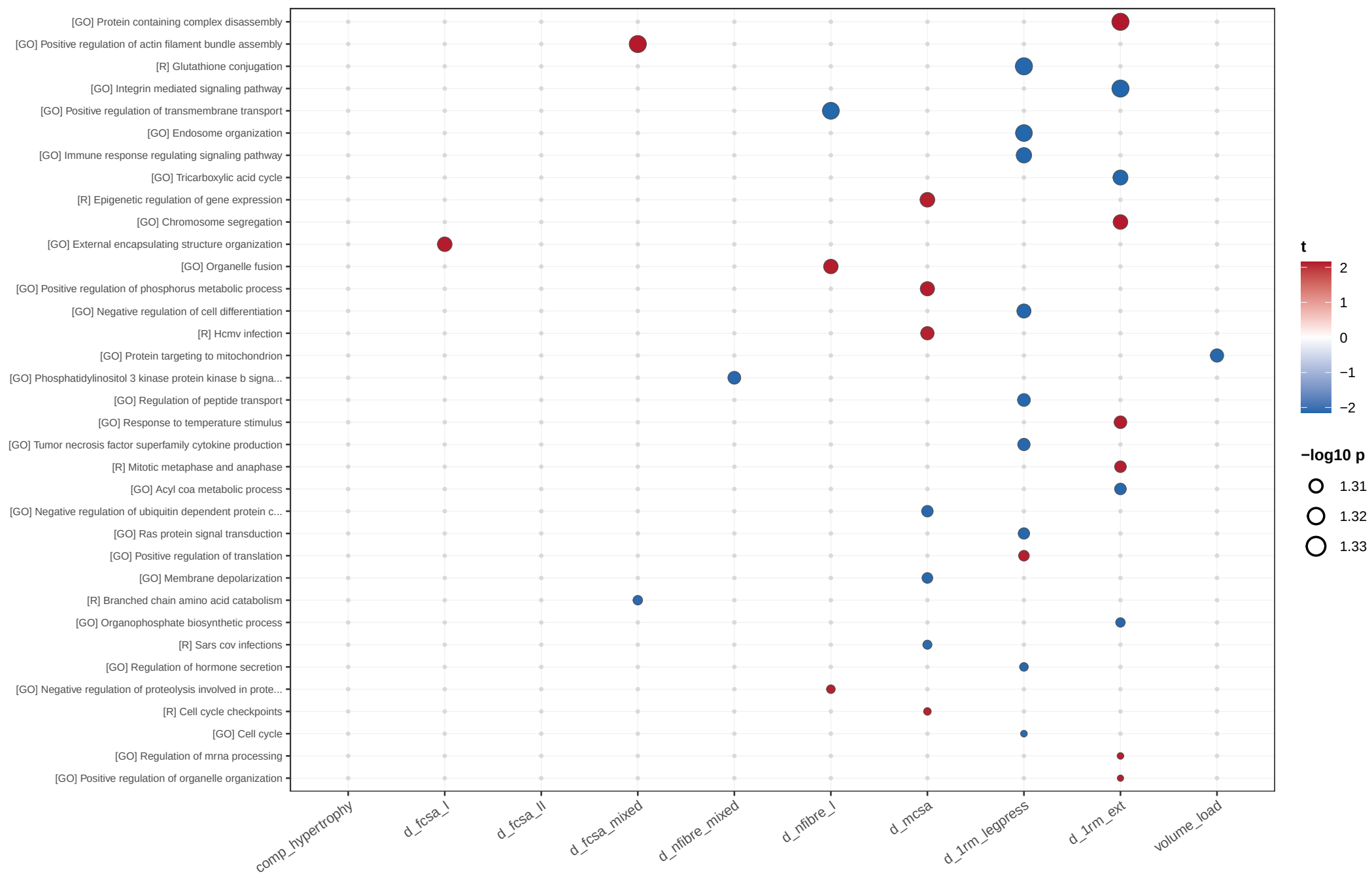
limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 376–450



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, level at T1 (7/7)

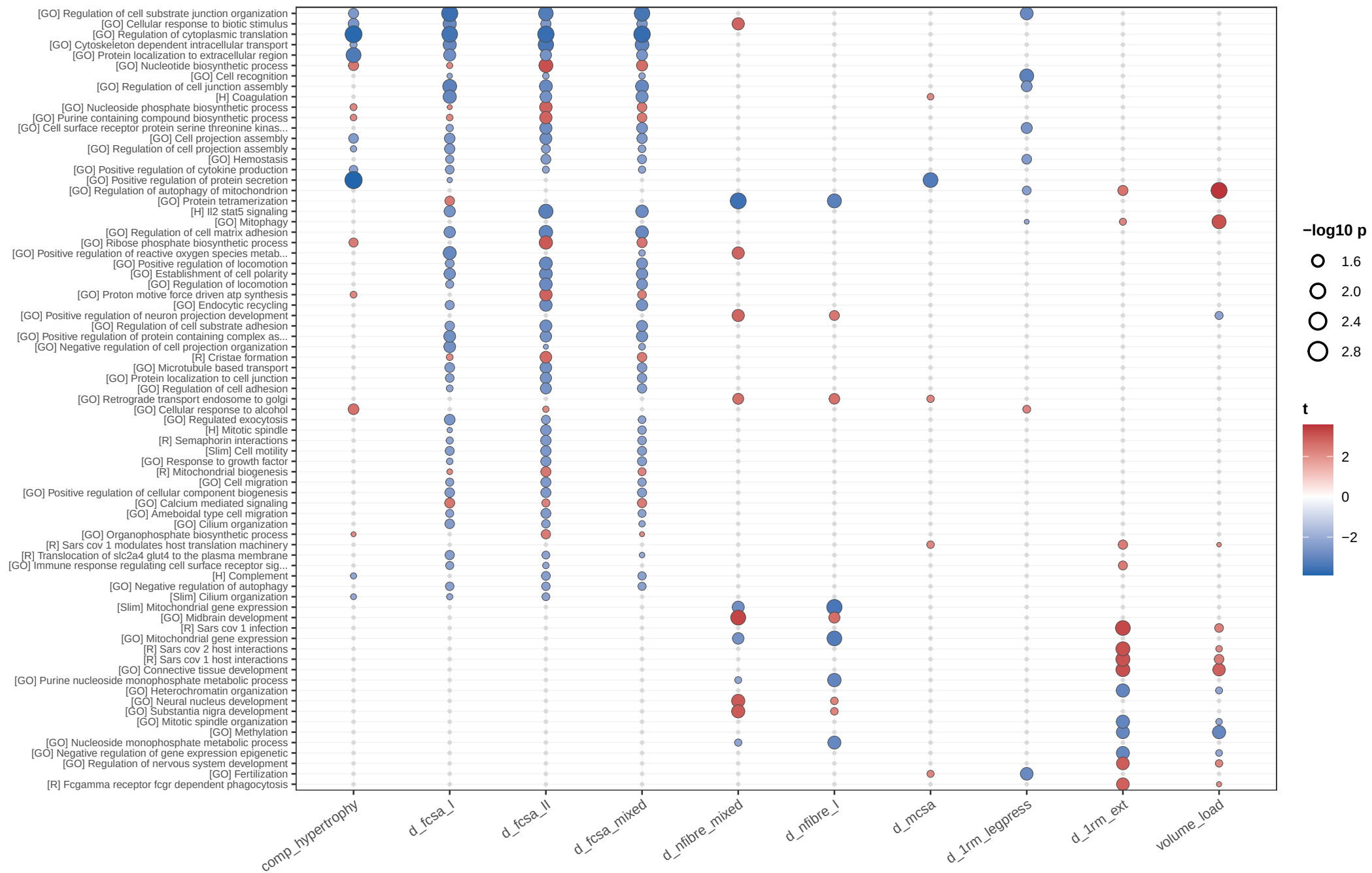
limma slope t against each phenotype, one row per subject; 485 features nominal in at least one column; rows 451–485



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, training change (T2 – T1) (1/6)

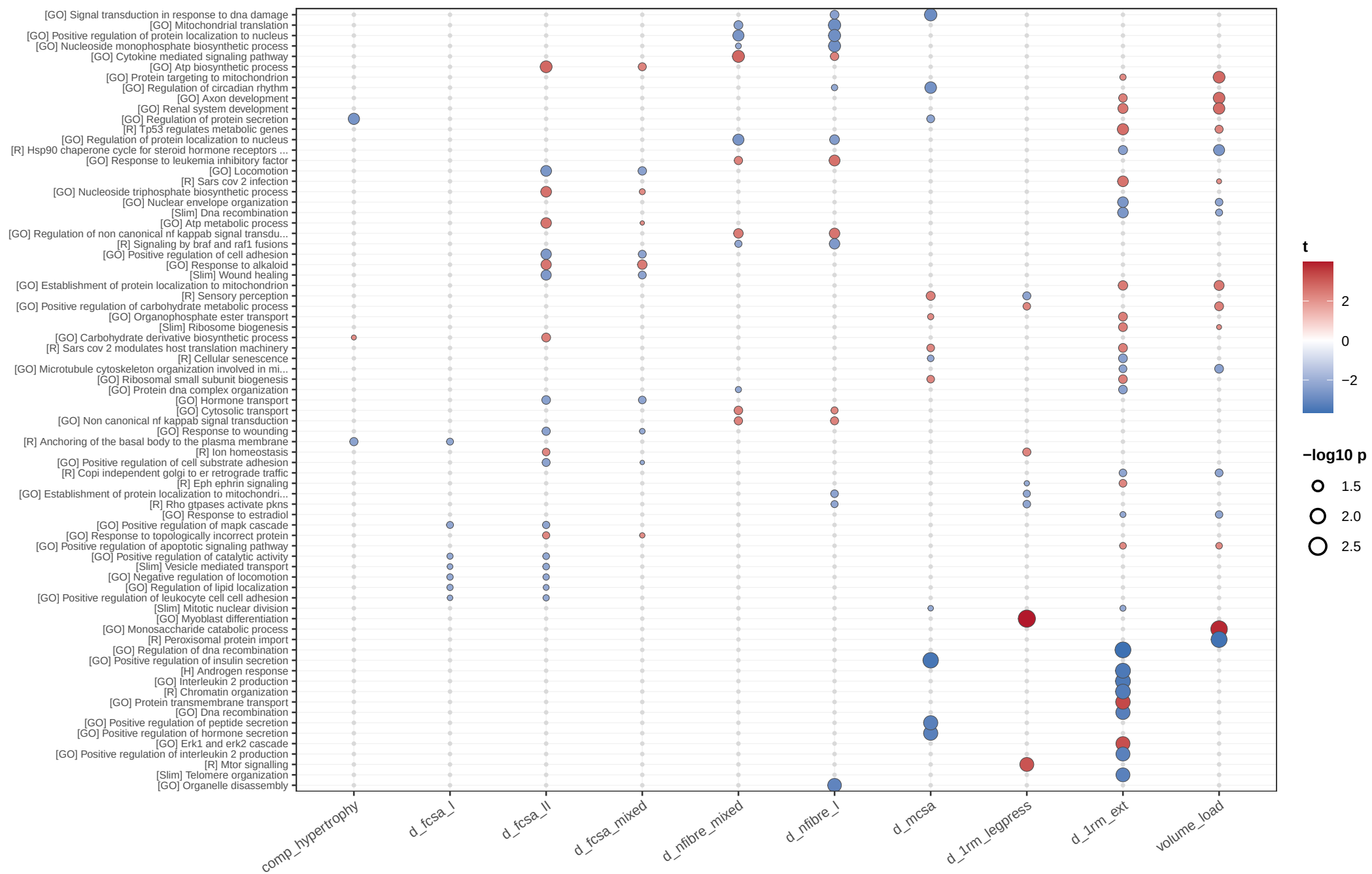
limma slope t against each phenotype, one row per subject; 407 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, training change (T2 – T1) (2/6)

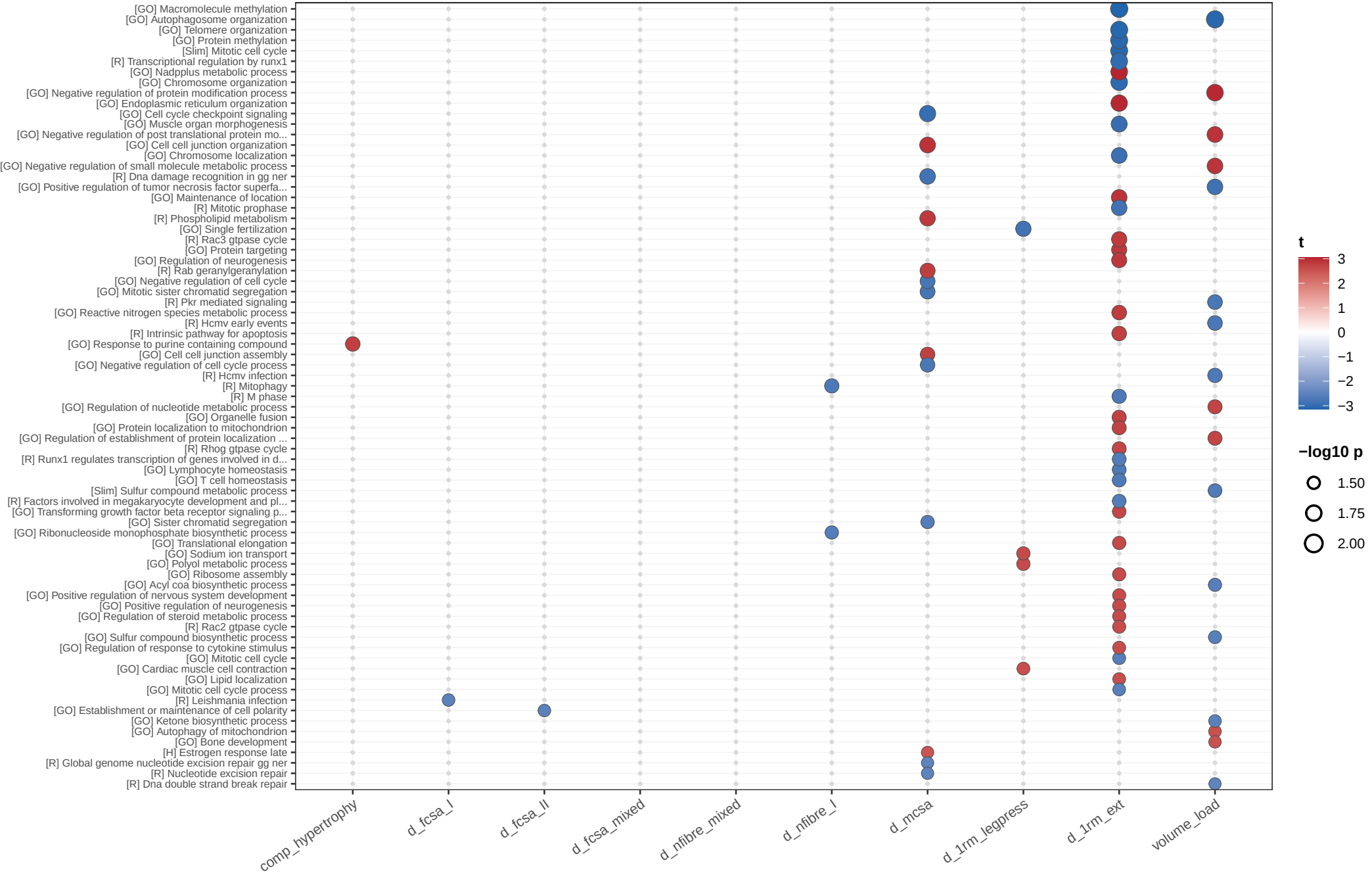
limma slope t against each phenotype, one row per subject; 407 features nominal in at least one column; rows 76–150



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, training change (T2 – T1) (3/6)

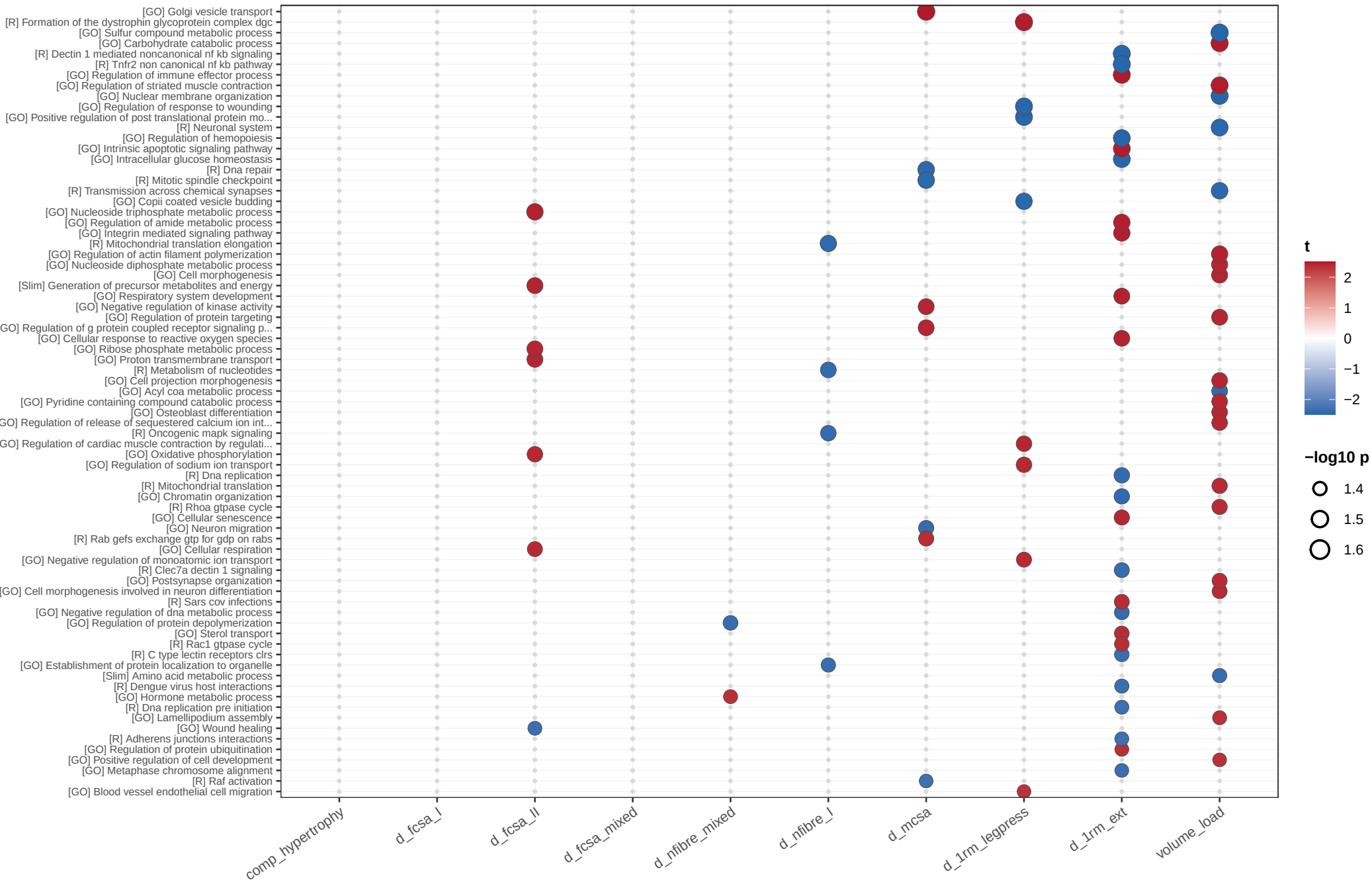
limma slope t against each phenotype, one row per subject; 407 features nominal in at least one column; rows 151–225



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, training change (T2 – T1) (4/6)

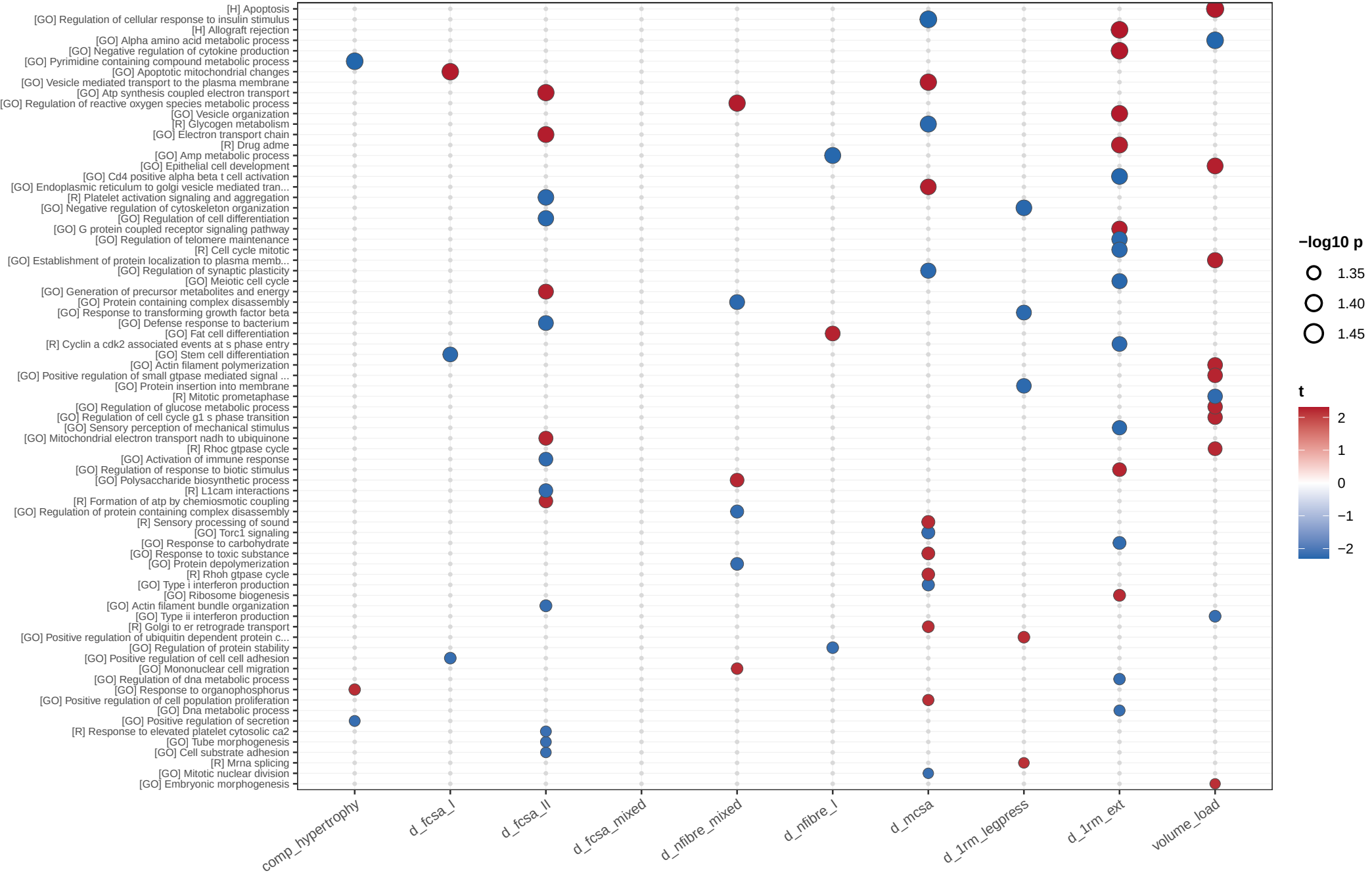
limma slope t against each phenotype, one row per subject; 407 features nominal in at least one column; rows 226–300



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, training change (T2 – T1) (5/6)

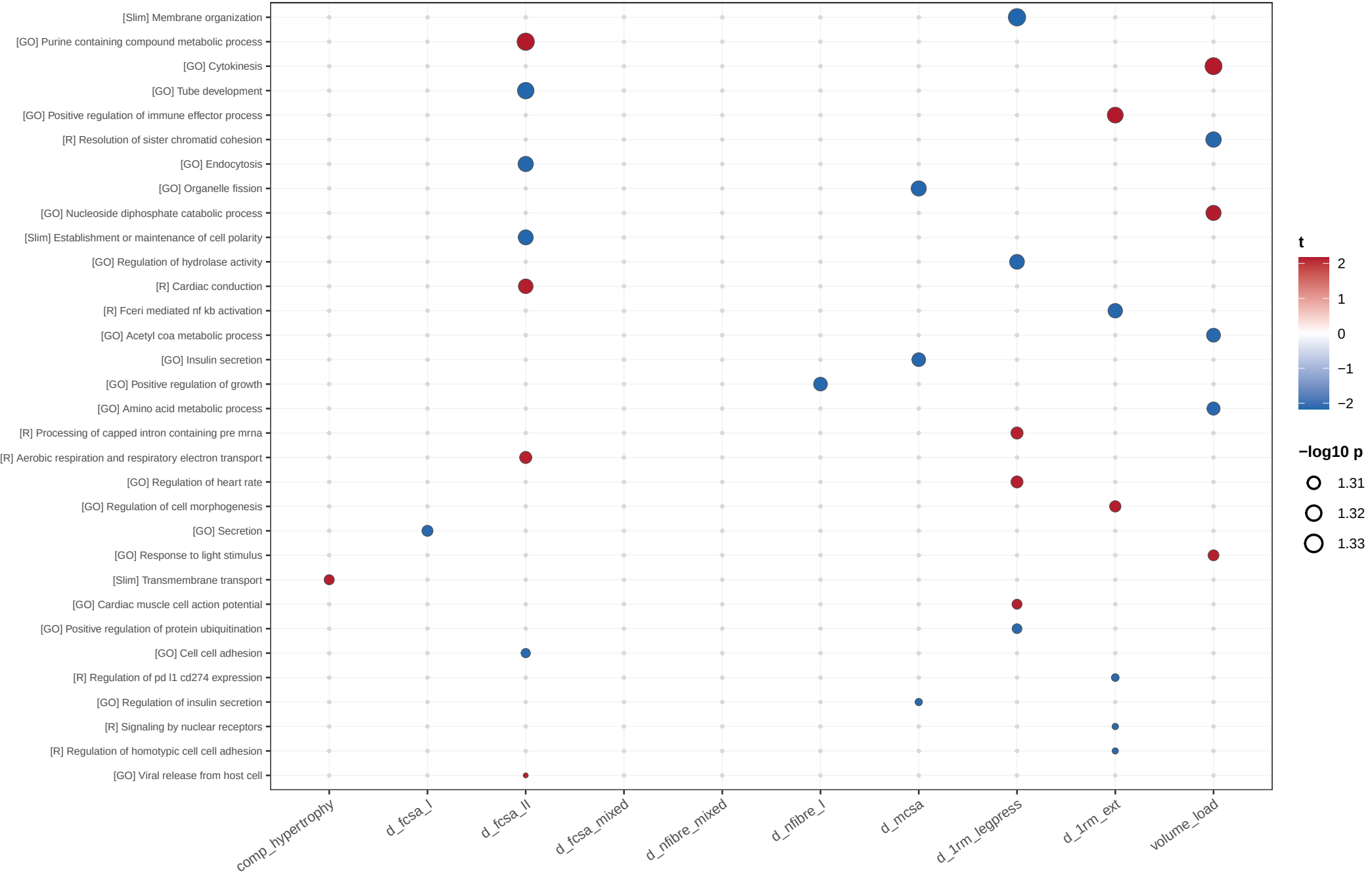
limma slope t against each phenotype, one row per subject; 407 features nominal in at least one column; rows 301–375



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, training change (T2 – T1) (6/6)

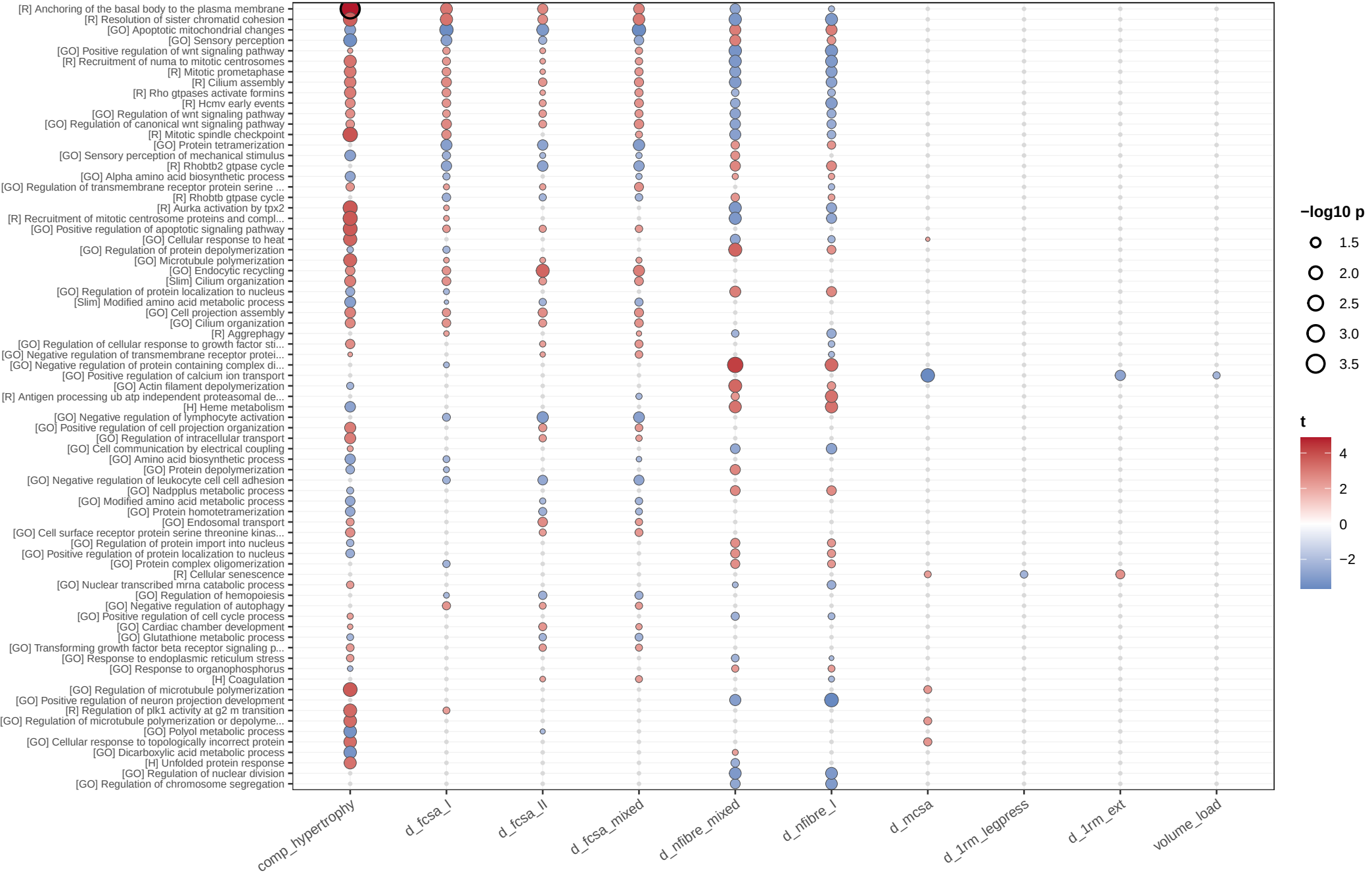
limma slope t against each phenotype, one row per subject; 407 features nominal in at least one column; rows 376–407



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, acute change (T3 – T2) (1/5)

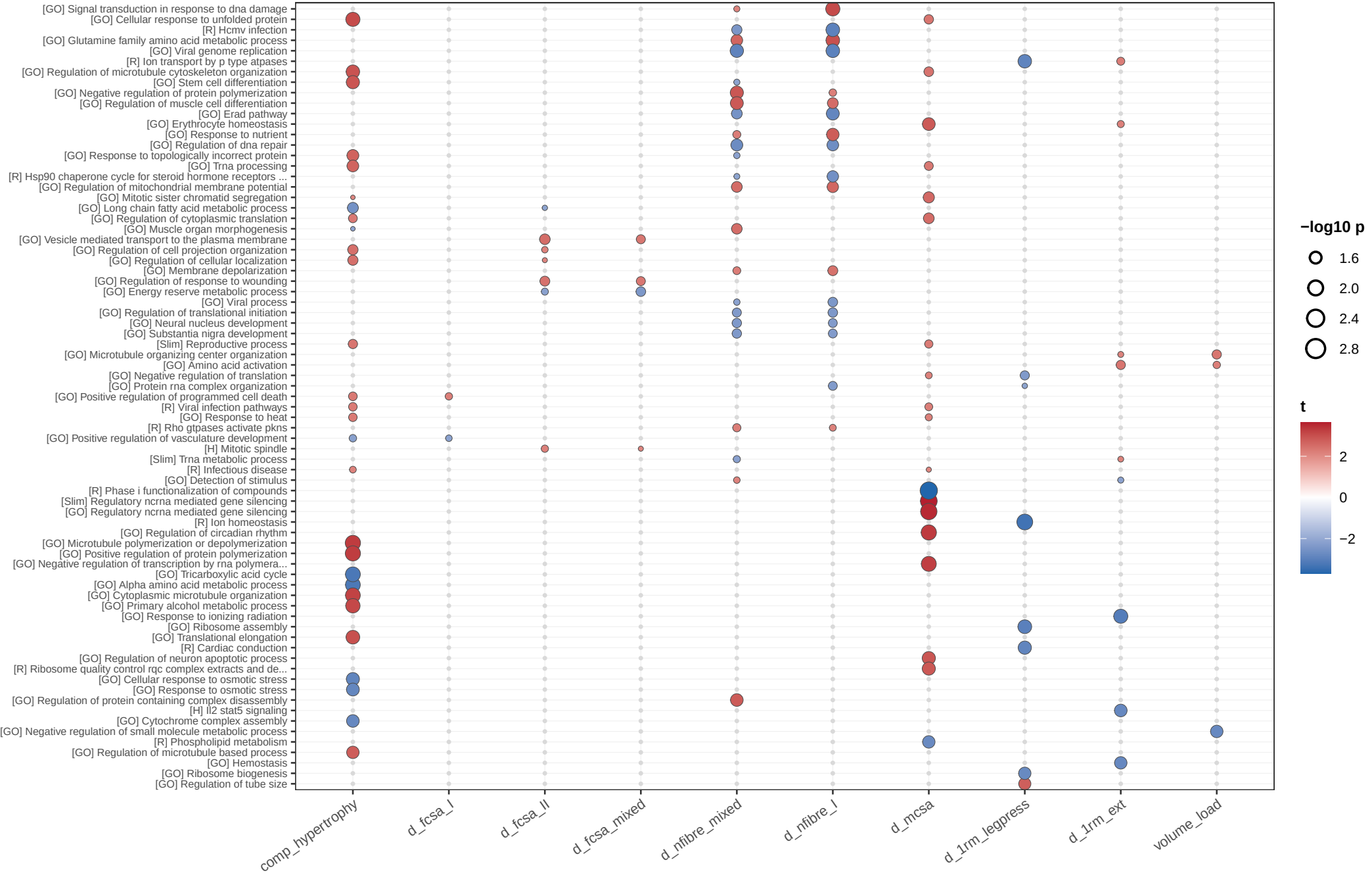
limma slope t against each phenotype, one row per subject; 366 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, acute change (T3 – T2) (2/5)

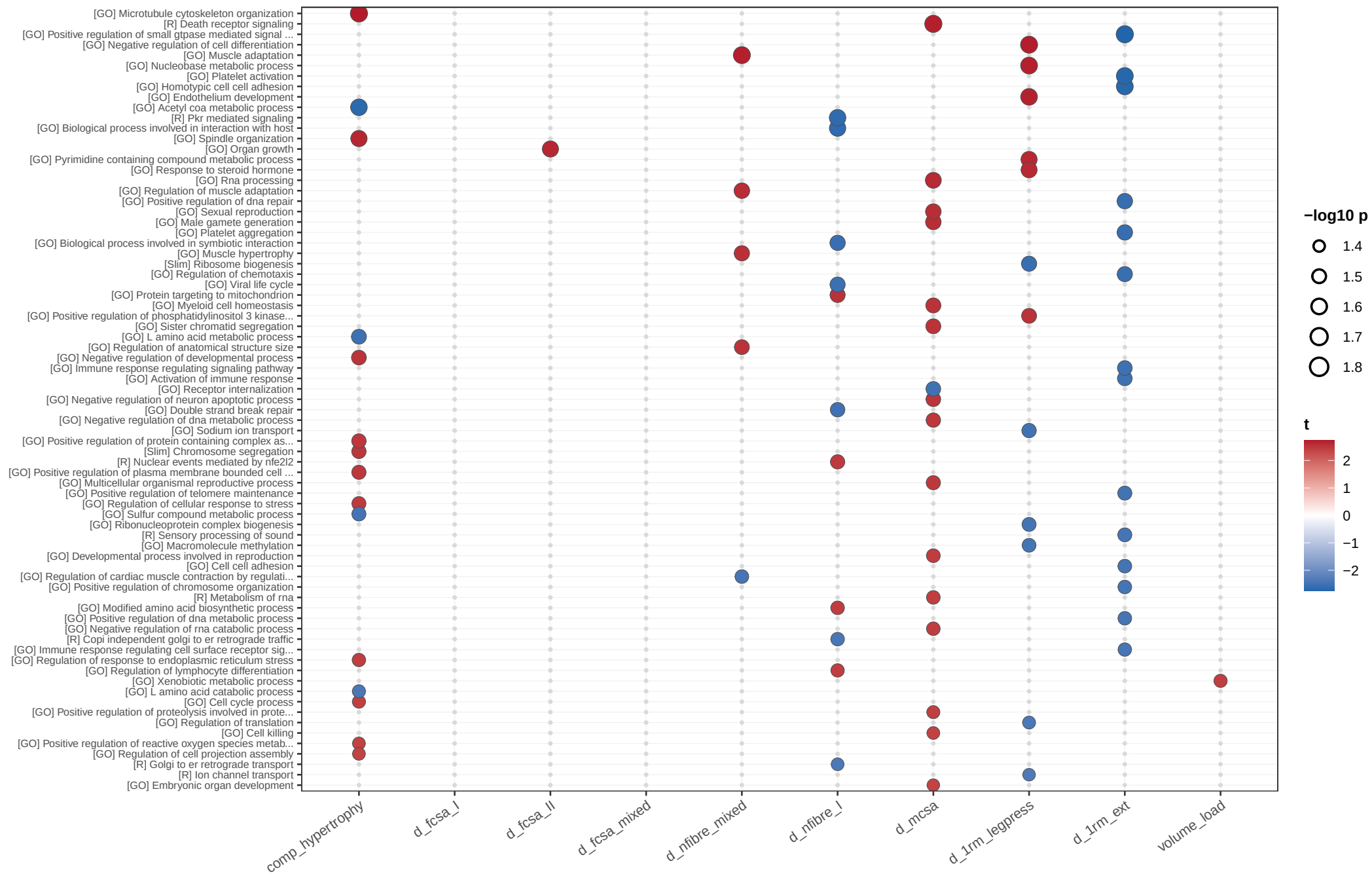
limma slope t against each phenotype, one row per subject; 366 features nominal in at least one column; rows 76–150



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = –log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, acute change (T3 – T2) (3/5)

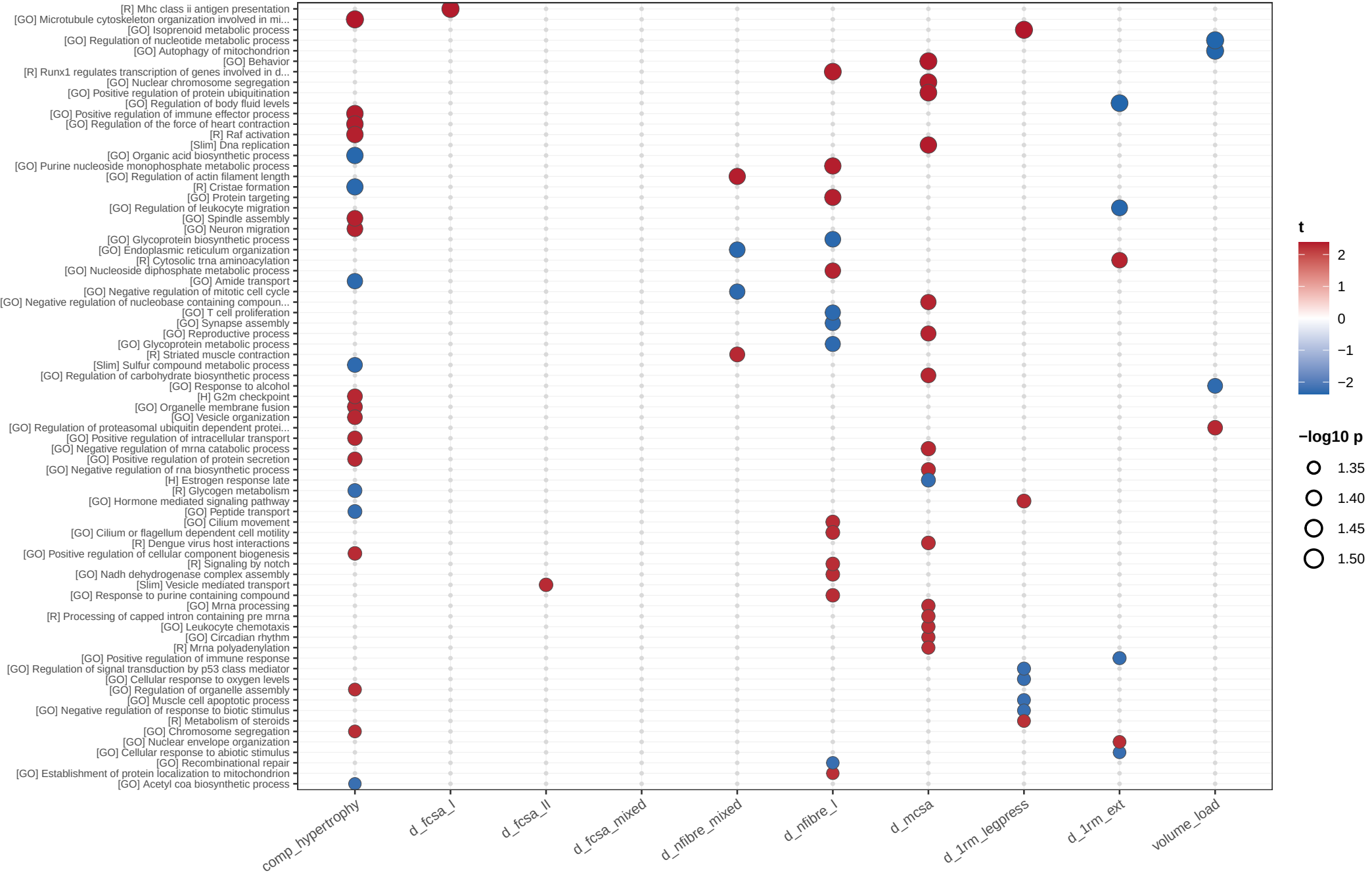
limma slope t against each phenotype, one row per subject; 366 features nominal in at least one column; rows 151–225



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, acute change (T3 – T2) (4/5)

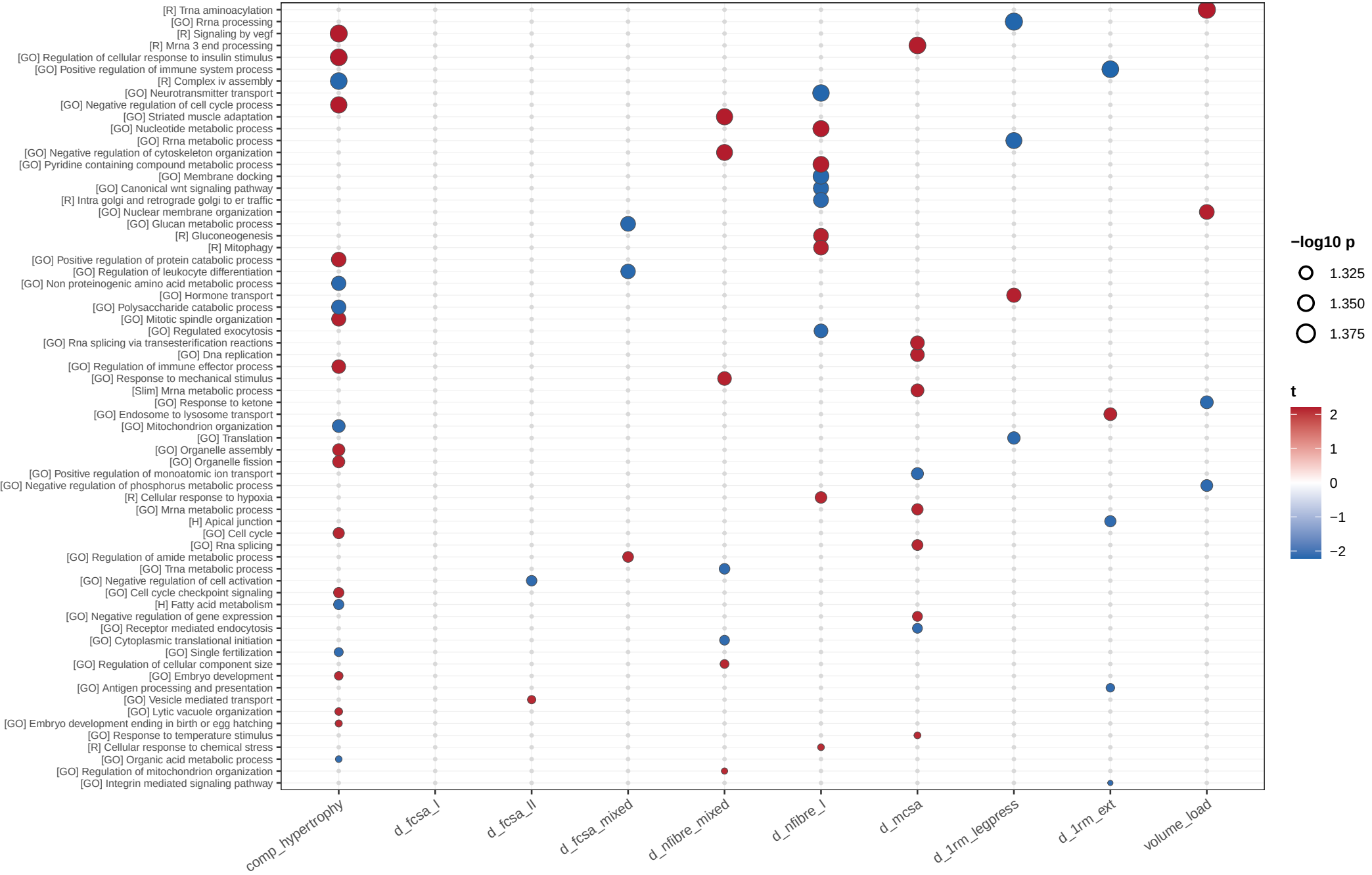
limma slope t against each phenotype, one row per subject; 366 features nominal in at least one column; rows 226–300



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.

Pathway association hits, acute change (T3 – T2) (5/5)

limma slope t against each phenotype, one row per subject; 366 features nominal in at least one column; rows 301–366



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 03_Pathways/04_Classify_Associate/c_data/pathway_screens.xlsx, sheet associate.